NACCA COURSES • MATHEMATICS

MATHS QUEST

BASIC 4

Explore 41 adventures in number, algebra, geometry, measurement and data.

41
LESSONS

71
CURRICULUM INDICATORS

615
ORIGINAL PRACTICE ITEMS

PRINTABLE LEARNER'S GUIDE · 2026 LEARNING EDITION

Your Maths Quest Starts Here

This book helps you think like a mathematician. Explore an example, try five guided Play tasks, then solve ten different Challenge problems.

LOOK & EXPLORE

Read the lesson goal, investigate its diagram and follow a worked example.

PLAY · 5 QUESTIONS

Use what you have just learned. Check your methods with a friend or adult.

CHALLENGE · 10 QUESTIONS

Try new numbers and fresh situations. Show your method, not only your answer.

REFLECT

Explain why your method works. Use the answer keys for self-checking after you try.

SAFETY AND SMART PRACTICE

Use clean counters, paper, pencils and a ruler. Ask an adult before going outside to measure, observe traffic, or handle tools. All activities can be done safely with drawings.

Find Your Next Adventure

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Find Your Next Adventure

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Building place value & Reading large numbers

LEARNING GOAL Building place value / Reading large numbers

MATHS MODEL · NUMBERS AND OPERATIONS

TEN THOUSANDS	THOUSANDS	HUNDREDS	TENS	ONES
2	3	4	0	5

$$23{,}405 = 20{,}000 + 3{,}000 + 400 + 5$$

LEARN THE IDEA

Learn to model number quantities, place value for multi-digit using graph sheets or multi-base materials up to 100,000 and read and write numbers in figures and in words up 100,000.

WORKED EXAMPLE 1

In 9,055, what is the value of the digit 5 in the ones place?

Answer: 5

The place value is 5 × 1 = 5.

WORKED EXAMPLE 2

Choose the words for 90,375.

Answer: ninety thousand three hundred and seventy-five

90,375 reads “ninety thousand three hundred and seventy-five”.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Building place value & Reading large numbers

PLAY · LEARN THE STEPS

1. **How many complete groups of 1 are in 14,682?**
   A) 14702 B) 14682 C) 14678 D) 14683
2. **Choose the words for 23,720.**
   A) twenty-three thousand eight hundred and twenty B) twenty-three thousand seven hundred and nineteen C) twenty-three thousand seven hundred and twenty D) twenty-three thousand seven hundred and thirty
3. **In 83,061, what is the value of the digit 0 in the hundreds place?**
   A) 2 B) 0 C) 5 D) 20
4. **Which number means “fifty-seven thousand nine hundred and twenty”?**
   A) 57,920 B) 57,910 C) 58,920 D) 58,020
5. **In 84,469, what is the value of the digit 4 in the thousands place?**
   A) 4,200 B) 4,050 C) 4,000 D) 4,002

CHALLENGE · DIFFERENT EXAMPLES

1. **A school has 72,215 bottle tops. How many complete bundles of 10 can it make?**
   A) 5 B) 7222 C) 7221 D) 7231
2. **How many complete groups of 100 are in 32,819?**
   A) 328 B) 332 C) 326 D) 323
3. **A place-value card shows 25,546. A learner covers all but the digit in the 10,000s column. What value does it show?**
   A) 20 B) 25,546 C) 2 D) 20,000
4. **In 80,015, what is the value of the digit 1 in the tens place?**
   A) 10 B) 15 C) 9 D) 7
5. **A school has 74,180 bottle tops. How many complete bundles of 10 can it make?**
   A) 0 B) 7419 C) 7418 D) 7428

Building place value & Reading large numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **On an election tally sheet, 59,442 is written. Which digit is in the thousands column?**

A) 4 B) 2 C) 9 D) 5

7 **Choose the words for 25,416.**

A) twenty-five thousand four hundred and sixteen B) twenty-five thousand four hundred and twenty-six C) twenty-five thousand five hundred and sixteen D) twenty-five thousand four hundred and fifteen

8 **On an election tally sheet, 32,954 is written. Which digit is in the tens column?**

A) 5 B) 9 C) 4 D) 8

9 **Which number means “forty-three thousand six hundred and eighty-four”?**

A) 44,684 B) 43,674 C) 43,684 D) 43,784

10 **On an election tally sheet, 12,419 is written. Which digit is in the hundreds column?**

A) 4 B) 2 C) 7 D) 1

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 14682	2. twenty-three thousand seven hundred and twenty	3. 0
4. 57,920	5. 4,000	6. 7221
7. 328	8. 20,000	9. 10
10. 7418	11. 9	12. twenty-five thousand four hundred and sixteen
13. 5	14. 43,684	15. 4

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

The number chart & Comparing and ordering

LEARNING GOAL The number chart / Comparing and ordering

MATHS MODEL · NUMBERS AND OPERATIONS



Find the step · Move forward or backward · Compare

LEARN THE IDEA

Learn to identify numbers in different positions around a given number in a number chart and compare and order whole numbers up to 10,000 and represent comparisons using the symbols “<”, “=”, “>”.

WORKED EXAMPLE 1

On a number chart with steps of 1, what is directly after 1,177?

Answer: 1,178

Move forward one step of 1.

WORKED EXAMPLE 2

Complete: 173 ___ 135.

Answer: >

Compare from the highest place value first.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

The number chart & Comparing and ordering

PLAY · LEARN THE STEPS

1. **On a number chart with steps of 1,000, what is directly after 1,270?**
   A) 2,273 B) 2,470 C) 2,267 D) 2,270
2. **Complete: 4,426 ___ 4,426.**
   A) < B) = C) >
3. **On a number chart with steps of 1, what is directly before 5,905?**
   A) 5,904 B) 5,924 C) 5,894 D) 5,901
4. **Complete: 5,396 ___ 5,358.**
   A) = B) > C) <
5. **On a number chart with steps of 100, what is directly after 4,611?**
   A) 4,711 B) 4,713 C) 4,708 D) 4,511

CHALLENGE · DIFFERENT EXAMPLES

1. **Kofi is on box 3,545 of a number chart counting by 10. What box comes next?**
   A) 3,535 B) 3,565 C) 3,555 D) 3,546
2. **On a number chart with steps of 10, what is directly after 6,422?**
   A) 6,412 B) 6,432 C) 6,428 D) 6,437
3. **Kofi is on box 3,423 of a number chart counting by 1. What box comes next?**
   A) 3,424 B) 3,422 C) 3,425 D) 3426
4. **On a number chart with steps of 100, what is directly after 5,931?**
   A) 6,051 B) 6,035 C) 6,031 D) 6,029
5. **Kofi is on box 4,768 of a number chart counting by 1. What box comes next?**
   A) 4771 B) 4,769 C) 4,770 D) 4,767

The number chart & Comparing and ordering

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two farms harvested 6,793 and 6,893 oranges. Which number is larger?**
A) 6,893 B) 6,793 C) 6,993 D) 6,693

7. **Complete: 6,013 ___ 5,773.**
A) < B) > C) =

8. **Two farms harvested 4,053 and 4,153 oranges. Which number is larger?**
A) 4,253 B) 4,053 C) 3,953 D) 4,153

9. **Complete: 3,038 ___ 3,608.**
A) < B) = C) >

10. **Two farms harvested 6,882 and 6,892 oranges. Which number is larger?**
A) 6,992 B) 6,782 C) 6,892 D) 6,882

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 2,270	2. =	3. 5,904
4. >	5. 4,711	6. 3,555
7. 6,432	8. 3,424	9. 6,031
10. 4,769	11. 6,893	12. >
13. 4,153	14. <	15. 6,892

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Rounding numbers & Skip counting

LEARNING GOAL Rounding numbers / Skip counting

MATHS MODEL · NUMBERS AND OPERATIONS



Find the step · Move forward or backward · Compare

LEARN THE IDEA

Learn to round (off, up, down) whole numbers up to 10000 to the nearest, thousands, hundreds and tens and skip count forwards and backwards in 50s and 100s up to and from 10000.

WORKED EXAMPLE 1

Round down to the number 1,107 to the 10.

Answer: 1,100

Use groups of 10 and the instruction “round down to”.

WORKED EXAMPLE 2

Skip count forwards in 50s from 8,950. What is the 3th number after the starting number?

Answer: 9,100

Apply +50 exactly 3 times.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Rounding numbers & Skip counting

PLAY · LEARN THE STEPS

1. **Round 3,116 up to the next multiple of 10.**
   A) 3,120 B) 3,130 C) 3,110
2. **Skip count backwards in 100s from 7,400. What is the 2nd number after the starting number?**
   A) 7,203 B) 7,200 C) 7,204 D) 7,199
3. **Round 7,489 down to the previous multiple of 1,000.**
   A) 8,000 B) 7,000 C) 6,000
4. **Skip count forwards in 50s from 3,300. What is the 3rd number after the starting number?**
   A) 4,450 B) 3,550 C) 3,440 D) 3,450
5. **Round 7,556 to the nearest 10.**
   A) 7,570 B) 7,560 C) 7,550

CHALLENGE · DIFFERENT EXAMPLES

1. **A market records about 8,695 visitors. Estimate to the nearest 100.**
   A) 8,695 B) 8,700 C) 8,800 D) 8,600
2. **Round 1,735 up to the next multiple of 100.**
   A) 1,900 B) 1,700 C) 1,800
3. **A market records about 8,626 visitors. Estimate to the nearest 1,000.**
   A) 8,626 B) 10,000 C) 9,000 D) 8,000
4. **Round 9,357 to the nearest 10.**
   A) 9,370 B) 9,360 C) 9,350
5. **A market records about 4,392 visitors. Estimate to the nearest 100.**
   A) 4,500 B) 4,300 C) 4,392 D) 4,400

Rounding numbers & Skip counting

CHALLENGE · APPLY WHAT YOU KNOW

6. **A counting game moves backward in 100s from 3,000. Where is the third landing?**
   A) 2,800 B) 2,900 C) 3,300 D) 2,700

7. **A counting game moves backward in 50s from 3,400. Where is the third landing?**
   A) 3,550 B) 3,300 C) 3,250 D) 3,350

8. **Skip count backwards in 50s from 3,050. What is the 1st number after the starting number?**
   A) 2,999 B) 3,000 C) 3,002 D) 3,020

9. **A counting game moves backward in 50s from 6,800. Where is the third landing?**
   A) 6,950 B) 6,700 C) 6,650 D) 6,750

10. **Skip count forwards in 100s from 8,600. What is the 2nd number after the starting number?**
    A) 8,800 B) 8,810 C) 8,798 D) 8,801

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 3,120	2. 7,200	3. 7,000
4. 3,450	5. 7,560	6. 8,700
7. 1,800	8. 9,000	9. 9,360
10. 4,400	11. 2,700	12. 3,250
13. 3,000	14. 6,650	15. 8,800

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Discover Roman numerals & Convert Roman numerals

LEARNING GOAL Discover Roman numerals / Convert Roman numerals

MATHS MODEL · NUMBERS AND OPERATIONS

I	V	X	XX	XXX
1	5	10	20	30

LEARN THE IDEA

Learn to develop an understanding of roman numeral system up xxx (i.e. 30) and count and convert hindu arabic numerals to roman numerals up to 30 and vice versa.

WORKED EXAMPLE 1

Which Roman numeral represents 2?

Answer: II

I=1, V=5 and X=10; IV=4, IX=9.

WORKED EXAMPLE 2

Convert XXV to Hindu-Arabic numerals.

Answer: 25

Group Xs, Vs and Is; a smaller numeral before a larger one is subtracted.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Discover Roman numerals & Convert Roman numerals

PLAY · LEARN THE STEPS

1. **Which Roman numeral represents 1?**
   A) XXI B) VI C) I D) XXIV
2. **Write 24 using Roman numerals.**
   A) XXVII B) XXIV C) XXV D) XXIII
3. **Which Roman numeral represents 25?**
   A) XV B) I C) XXV D) XVI
4. **Convert XII to Hindu-Arabic numerals.**
   A) 7 B) 9 C) 12 D) 15
5. **Which Roman numeral represents 2?**
   A) VII B) XIX C) II D) IV

CHALLENGE · DIFFERENT EXAMPLES

1. **A scoreboard uses Roman numbers. Which symbol should appear for position 6?**
   A) XVI B) XII C) VI D) X
2. **A scoreboard uses Roman numbers. Which symbol should appear for position 14?**
   A) XIV B) XVIII C) XXIV D) XX
3. **A scoreboard uses Roman numbers. Which symbol should appear for position 21?**
   A) XXVII B) I C) XXV D) XXI
4. **A scoreboard uses Roman numbers. Which symbol should appear for position 29?**
   A) IX B) XXIX C) V D) III
5. **Which Roman numeral represents 7?**
   A) XVII B) XXII C) VII D) III

Discover Roman numerals & Convert Roman numerals

CHALLENGE · APPLY WHAT YOU KNOW

6. **A clock face is labelled XXVIII. What Hindu-Arabic number matches the label?**
A) 27 B) 30 C) 28 D) 2

7. **Convert II to Hindu-Arabic numerals.**
A) 0 B) 102 C) 12 D) 2

8. **A clock face is labelled XXX. What Hindu-Arabic number matches the label?**
A) 29 B) 30 C) 2 D) 4

9. **Write 6 using Roman numerals.**
A) VI B) VIII C) XVI D) XVIII

10. **A clock face is labelled I. What Hindu-Arabic number matches the label?**
A) 5 B) 6 C) 1 D) 3

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. I	2. XXIV	3. XXV
4. 12	5. II	6. VI
7. XIV	8. XXI	9. XXIX
10. VII	11. 28	12. 2
13. 30	14. VI	15. 1

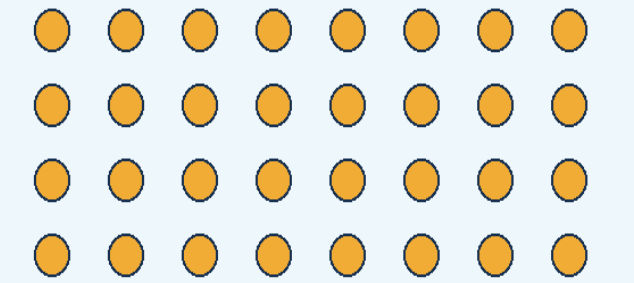
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Factors & Highest common factors

LEARNING GOAL Factors / Highest common factors

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to determine set of factors of a given numbers up to 50 and determine the highest common factor (hcf) of any two whole numbers between 1 and 50.

WORKED EXAMPLE 1

14 objects are put into groups of 7. How many groups are made?

Answer: 2

Factors make equal groups.

WORKED EXAMPLE 2

Find the highest common factor (HCF) of 8 and 23.

Answer: 1

The greatest number dividing both 8 and 23 is 1.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Factors & Highest common factors

PLAY · LEARN THE STEPS

1. **27 objects are put into groups of 9. How many groups are made?**
   A) 23 B) 8 C) 3 D) 1
2. **Find the highest common factor (HCF) of 9 and 39.**
   A) 103 B) 3 C) 1003 D) 5
3. **Which number is a factor of 29?**
   A) 29 B) 6 C) 2 D) 9
4. **Find the highest common factor (HCF) of 36 and 43.**
   A) 6 B) 1001 C) 2 D) 1
5. **20 objects are put into groups of 5. How many groups are made?**
   A) 5 B) 104 C) 9 D) 4

CHALLENGE · DIFFERENT EXAMPLES

1. **A gardener plants 18 seedlings in equal rows of 9. How many complete rows are needed?**
   A) 4 B) 18 C) 2 D) 9
2. **Which number is a factor of 47?**
   A) 5 B) 13 C) 47 D) 23
3. **A gardener plants 35 seedlings in equal rows of 7. How many complete rows are needed?**
   A) 35 B) 5 C) 10 D) 7
4. **10 objects are put into groups of 5. How many groups are made?**
   A) 1 B) 2 C) 7 D) 102
5. **A gardener plants 33 seedlings in equal rows of 11. How many complete rows are needed?**
   A) 5 B) 11 C) 33 D) 3

Factors & Highest common factors

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two pupils have 28 and 21 stickers. What is the largest equal-sized packet each can make without leftovers?**
   A) 9 B) 7 C) 8 D) 3

7. **Two pupils have 27 and 36 stickers. What is the largest equal-sized packet each can make without leftovers?**
   A) 10 B) 11 C) 4 D) 9

8. **Find the highest common factor (HCF) of 31 and 9.**
   A) 3 B) 4 C) 1001 D) 1

9. **Two pupils have 40 and 24 stickers. What is the largest equal-sized packet each can make without leftovers?**
   A) 8 B) 4 C) 10 D) 9

10. **Find the highest common factor (HCF) of 8 and 12.**
    A) 4 B) 5 C) 204 D) 9

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 3	2. 3	3. 29
4. 1	5. 4	6. 2
7. 47	8. 5	9. 2
10. 3	11. 7	12. 9
13. 1	14. 8	15. 4

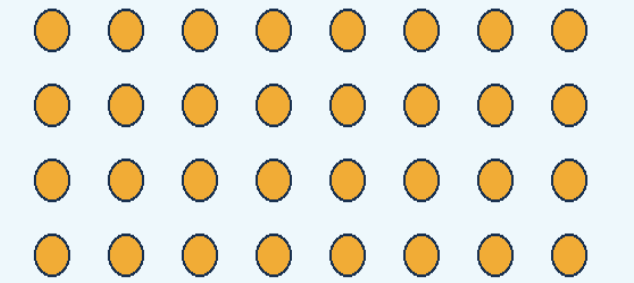
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Lowest common multiples & Factors and multiples

LEARNING GOAL Lowest common multiples / Factors and multiples

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to determine the lowest common multiple (lcm) of at least any two given numbers up to 100 and recognise the relationship between factors and multiples.

WORKED EXAMPLE 1

Find the lowest common multiple (LCM) of 2 and 2.

Answer: 2

2 is the smallest positive multiple of both numbers.

WORKED EXAMPLE 2

2 is a factor of 6. Which statement explains their relationship?

Answer: 6 is a multiple of 2

6 = 2 × 3.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Lowest common multiples & Factors and multiples

PLAY · LEARN THE STEPS

1. **Find the lowest common multiple (LCM) of 8 and 13.**
   A) 102 B) 104 C) 107 D) 106
2. **5 is a factor of 30. Which statement explains their relationship?**
   A) 5 is not a factor of 30 B) 30 is less than 5 C) 30 is a multiple of 5 D) 5 and 30 have no related factors
3. **Find the lowest common multiple (LCM) of 2 and 11.**
   A) 2 B) 22 C) 26 D) 18
4. **4 is a factor of 20. Which statement explains their relationship?**
   A) 4 and 20 have no related factors B) 20 is a multiple of 4 C) 20 is less than 4 D) 4 is not a factor of 20
5. **Find the lowest common multiple (LCM) of 8 and 6.**
   A) 24 B) 14 C) 20 D) 25

CHALLENGE · DIFFERENT EXAMPLES

1. **A bell rings every 4 minutes and a drum every 5 minutes. Starting together, after how many minutes will they next sound together?**
   A) 20 B) 21 C) 9 D) 2
2. **Find the lowest common multiple (LCM) of 4 and 9.**
   A) 36 B) 56 C) 86 D) 1036
3. **A bell rings every 2 minutes and a drum every 8 minutes. Starting together, after how many minutes will they next sound together?**
   A) 10 B) 8 C) 17 D) 7
4. **Find the lowest common multiple (LCM) of 8 and 2.**
   A) 58 B) 208 C) 4 D) 8
5. **A bell rings every 8 minutes and a drum every 5 minutes. Starting together, after how many minutes will they next sound together?**
   A) 13 B) 4 C) 41 D) 40

Lowest common multiples & Factors and multiples

CHALLENGE · APPLY WHAT YOU KNOW

6 **A shop packs 56 pencils in groups of 8. Which relationship is true?**
A) 56 is a multiple of 8 B) 56 is a factor of 8 C) 56 is smaller than 8 D) 8 is not a factor of 56

7 **5 is a factor of 15. Which statement explains their relationship?**
A) 15 is a multiple of 5 B) 5 is not a factor of 15 C) 15 is less than 5 D) 5 and 15 have no related factors

8 **A shop packs 14 pencils in groups of 7. Which relationship is true?**
A) 14 is a factor of 7 B) 14 is a multiple of 7 C) 7 is not a factor of 14 D) 14 is smaller than 7

9 **7 is a factor of 35. Which statement explains their relationship?**
A) 35 is less than 7 B) 35 is a multiple of 7 C) 7 is not a factor of 35 D) 7 and 35 have no related factors

10 **A shop packs 40 pencils in groups of 8. Which relationship is true?**
A) 40 is smaller than 8 B) 40 is a factor of 8 C) 8 is not a factor of 40 D) 40 is a multiple of 8

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 104	2. 30 is a multiple of 5	3. 22
4. 20 is a multiple of 4	5. 24	6. 20
7. 36	8. 8	9. 8
10. 40	11. 56 is a multiple of 8	12. 15 is a multiple of 5
13. 14 is a multiple of 7	14. 35 is a multiple of 7	15. 40 is a multiple of 8

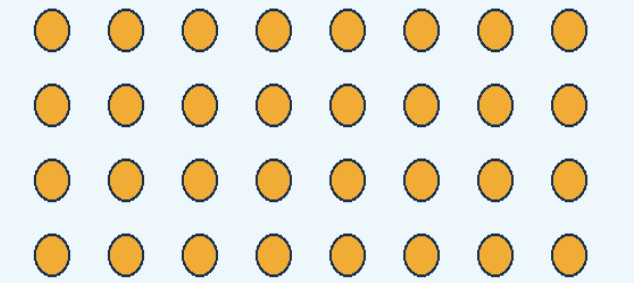
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Square-number patterns & Square arrays

LEARNING GOAL Square-number patterns / Square arrays

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to generate and analyse patterns in square numbers and represent square numbers using factors.

WORKED EXAMPLE 1

Continue the square-number sequence: 9, 16, ___.

Answer: 25

Square numbers are 1×1, 2×2, 3×3, ...

WORKED EXAMPLE 2

An array has 2 rows and 2 columns. How many small squares are there?

Answer: 4

2 × 2 = 4.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Square-number patterns & Square arrays

PLAY · LEARN THE STEPS

1. **Continue the square-number sequence: 289, 324, ___.**
   A) 358 B) 365 C) 381 D) 361
2. **An array has 22 rows and 22 columns. How many small squares are there?**
   A) 484 B) 504 C) 488 D) 485
3. **Continue the square-number sequence: 256, 289, ___.**
   A) 320 B) 314 C) 1324 D) 324
4. **An array has 10 rows and 10 columns. How many small squares are there?**
   A) 96 B) 80 C) 100 D) 150
5. **Continue the square-number sequence: 441, 484, ___.**
   A) 529 B) 579 C) 509 D) 519

CHALLENGE · DIFFERENT EXAMPLES

1. **A pattern uses square tiles: 1, 4, 9, 16, ... Which number represents a 12 by 12 square?**
   A) 145 B) 24 C) 156 D) 144
2. **Continue the square-number sequence: 81, 100, ___.**
   A) 116 B) 121 C) 101 D) 118
3. **A pattern uses square tiles: 1, 4, 9, 16, ... Which number represents a 8 by 8 square?**
   A) 65 B) 16 C) 64 D) 72
4. **A pattern uses square tiles: 1, 4, 9, 16, ... Which number represents a 9 by 9 square?**
   A) 82 B) 81 C) 90 D) 18
5. **A pattern uses square tiles: 1, 4, 9, 16, ... Which number represents a 14 by 14 square?**
   A) 196 B) 28 C) 210 D) 197

Square-number patterns & Square arrays

CHALLENGE · APPLY WHAT YOU KNOW

6 **A school garden is laid out in 6 rows of 6 seedlings. How many seedlings altogether?**

A) 42 B) 12 C) 35 D) 36

7 **An array has 7 rows and 7 columns. How many small squares are there?**

A) 249 B) 49 C) 51 D) 52

8 **A school garden is laid out in 13 rows of 13 seedlings. How many seedlings altogether?**

A) 168 B) 26 C) 182 D) 169

9 **An array has 17 rows and 17 columns. How many small squares are there?**

A) 289 B) 279 C) 287 D) 288

10 **A school garden is laid out in 4 rows of 4 seedlings. How many seedlings altogether?**

A) 20 B) 8 C) 16 D) 15

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 361	2. 484	3. 324
4. 100	5. 529	6. 144
7. 121	8. 64	9. 81
10. 196	11. 36	12. 49
13. 169	14. 289	15. 16

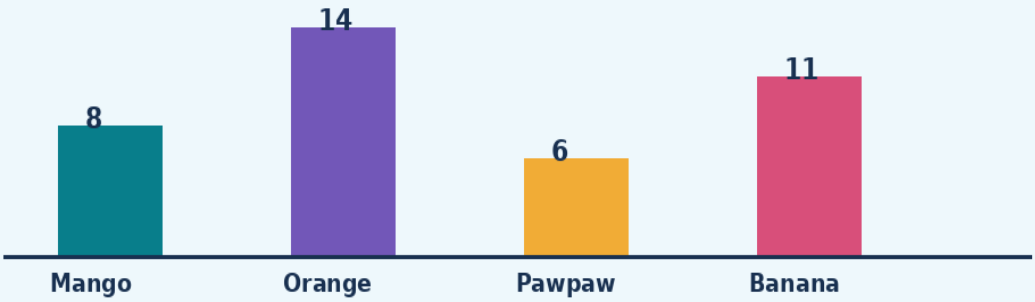
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Negative numbers around us & Crossing zero on a number line

LEARNING GOAL Negative numbers around us / Crossing zero on a number line

MATHS MODEL · NUMBERS AND OPERATIONS



LEARN THE IDEA

Learn to describe real life situations using positive and negative values and count forwards and backwards with positive and negative whole numbers through zero.

WORKED EXAMPLE 1

The temperature is -11°C. It becomes 1°C colder. What is the new temperature?

Answer: -12°C

Colder means subtract 1.

WORKED EXAMPLE 2

On a number line, start at 9. Move 6 steps right. Where do you land?

Answer: 15

Right increases; left decreases.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Negative numbers around us & Crossing zero on a number line

PLAY · LEARN THE STEPS

1. **The temperature is 4°C. It becomes 2°C colder. What is the new temperature?**
   A) 4°C B) 0°C C) -2°C D) 2°C
2. **On a number line, start at -11. Move 6 steps left. Where do you land?**
   A) 33 B) 83 C) 3 D) -17
3. **The temperature is 11°C. It becomes 9°C warmer. What is the new temperature?**
   A) 18°C B) 22°C C) 20°C D) -20°C
4. **On a number line, start at 1. Move 5 steps right. Where do you land?**
   A) 8 B) 11 C) 26 D) 6
5. **The temperature is 6°C. It becomes 5°C colder. What is the new temperature?**
   A) 1°C B) 3°C C) -1°C

CHALLENGE · DIFFERENT EXAMPLES

1. **A mountain village is at 9°C. The reading changes by −3°C. What temperature is recorded?**
   A) 6°C B) 8°C C) 9°C D) 4°C
2. **The temperature is 7°C. It becomes 10°C warmer. What is the new temperature?**
   A) -17°C B) 19°C C) 17°C D) 15°C
3. **A mountain village is at 1°C. The reading changes by +4°C. What temperature is recorded?**
   A) 7°C B) 1°C C) 3°C D) 5°C
4. **The temperature is 4°C. It becomes 4°C colder. What is the new temperature?**
   A) -2°C B) 0°C C) 2°C
5. **A mountain village is at -12°C. The reading changes by −10°C. What temperature is recorded?**
   A) -12°C B) -22°C C) -20°C D) -24°C

Negative numbers around us & Crossing zero on a number line

CHALLENGE · APPLY WHAT YOU KNOW

6 **A number-line counter begins at 3. It travels 6 spaces left. What number does it reach?**

A) -5 B) 3 C) -3 D) -2

7 **On a number line, start at 6. Move 1 step right. Where do you land?**

A) 5 B) 207 C) 7 D) 11

8 **A number-line counter begins at -10. It travels 10 spaces right. What number does it reach?**

A) 1 B) -2 C) -10 D) 0

9 **On a number line, start at -2. Move 6 steps left. Where do you land?**

A) 992 B) -8 C) -1008 D) 12

10 **A number-line counter begins at -10. It travels 3 spaces left. What number does it reach?**

A) -12 B) -10 C) -15 D) -13

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 2°C	2. -17	3. 20°C
4. 6	5. 1°C	6. 6°C
7. 17°C	8. 5°C	9. 0°C
10. -22°C	11. -3	12. 7
13. 0	14. -8	15. -13

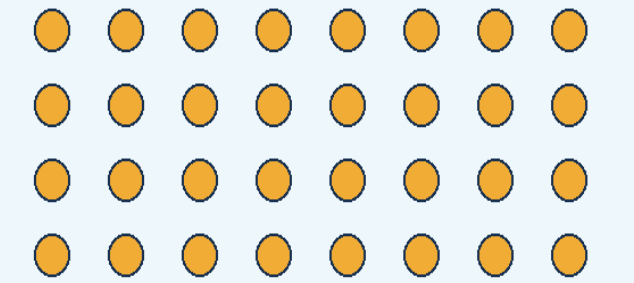
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplication facts

LEARNING GOAL Multiplication facts

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to recall multiplication facts through the tables up to 11 × 11.

WORKED EXAMPLE 1

Calculate 2 × 2.

Answer: 4

2 groups of 2 give 4.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Multiplication facts

PLAY · LEARN THE STEPS

1. **Calculate 4 × 10.**
   A) 40 B) 1040 C) 140 D) 90
2. **Calculate 8 × 5.**
   A) 43 B) 60 C) 40 D) 42
3. **Calculate 3 × 3.**
   A) 9 B) 12 C) 11 D) 6
4. **Calculate 3 × 9.**
   A) 25 B) 127 C) 26 D) 27
5. **Calculate 5 × 7.**
   A) 55 B) 33 C) 35 D) 235

CHALLENGE · DIFFERENT EXAMPLES

1. **There are 4 trays with 4 oranges in each. How many oranges are there altogether?**
   A) 8 B) 12 C) 20 D) 16
2. **There are 6 rows with 10 chairs in each. How many chairs are there altogether?**
   A) 16 B) 70 C) 50 D) 60
3. **Calculate 5 × 5.**
   A) 15 B) 21 C) 26 D) 25
4. **There are 4 packets with 10 pencils in each. How many pencils are there altogether?**
   A) 14 B) 50 C) 30 D) 40
5. **There are 12 teams with 4 players in each. How many players are there altogether?**
   A) 48 B) 16 C) 52 D) 44

Multiplication facts

CHALLENGE · APPLY WHAT YOU KNOW

6. **Calculate 7 × 9.**
A) 68 B) 53 C) 63 D) 61

7. **There are 11 tables with 11 books in each. How many books are there altogether?**
A) 22 B) 132 C) 110 D) 121

8. **Calculate 4 × 4.**
A) 15 B) 21 C) 12 D) 16

9. **There are 7 crates with 8 mangoes in each. How many mangoes are there altogether?**
A) 56 B) 64 C) 15 D) 48

10. **Calculate 8 × 4.**
A) 22 B) 31 C) 32 D) 34

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 40	2. 40	3. 9
4. 27	5. 35	6. 16
7. 60	8. 25	9. 40
10. 48	11. 63	12. 121
13. 16	14. 56	15. 32

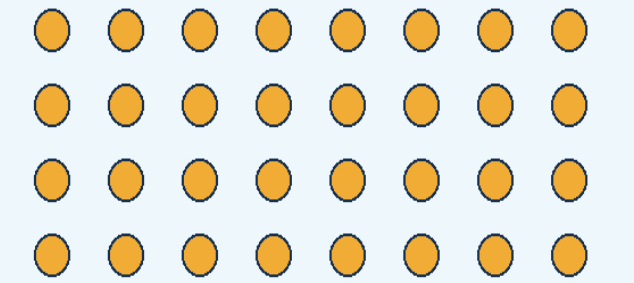
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Doubling as a strategy & Mental multiplication

LEARNING GOAL Doubling as a strategy / Mental multiplication

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to apply mental mathematics strategies and number properties, such as skip counting from a known fact using doubling or halving using patterns in the 9s facts using repeated doubling or halving to determine answers for basic multiplication facts to 81 and related division facts and …

WORKED EXAMPLE 1

You know 2 × 3 = 6. What is 4 × 3 by doubling?

Answer: 12

Double 6 to get 12.

WORKED EXAMPLE 2

Multiply 14 × 10 mentally using place value.

Answer: 140

Think 14 × 1, then multiply by 10.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Doubling as a strategy & Mental multiplication

PLAY · LEARN THE STEPS

1. **You know 4 × 3 = 12. What is 8 × 3 by doubling?**
   A) 74 B) 24 C) 25 D) 27
2. **Multiply 4 × 50 mentally using place value.**
   A) 210 B) 150 C) 200 D) 250
3. **You know 9 × 9 = 81. What is 18 × 9 by doubling?**
   A) 167 B) 164 C) 162 D) 157
4. **Multiply 12 × 50 mentally using place value.**
   A) 605 B) 600 C) 596 D) 620
5. **You know 9 × 4 = 36. What is 18 × 4 by doubling?**
   A) 272 B) 72 C) 92 D) 75

CHALLENGE · DIFFERENT EXAMPLES

1. **8 baskets hold 7 tomatoes each. Twice as many baskets are filled the same way. How many tomatoes are there?**
   A) 112 B) 119 C) 56 D) 105
2. **You know 6 × 10 = 60. What is 12 × 10 by doubling?**
   A) 220 B) 100 C) 125 D) 120
3. **4 baskets hold 8 tomatoes each. Twice as many baskets are filled the same way. How many tomatoes are there?**
   A) 32 B) 64 C) 72 D) 56
4. **You know 2 × 5 = 10. What is 4 × 5 by doubling?**
   A) 10 B) 16 C) 20 D) 25
5. **You know 5 × 6 = 30. What is 10 × 6 by doubling?**
   A) 60 B) 56 C) 260 D) 70

Doubling as a strategy & Mental multiplication

CHALLENGE · APPLY WHAT YOU KNOW

6. **There are 50 trays with 4 oranges in each. How many oranges are there altogether?**
   A) 200 B) 204 C) 196 D) 54

7. **Multiply 5 × 100 mentally using place value.**
   A) 501 B) 500 C) 450 D) 510

8. **There are 10 rows with 7 chairs in each. How many chairs are there altogether?**
   A) 77 B) 70 C) 63 D) 17

9. **Multiply 9 × 100 mentally using place value.**
   A) 897 B) 900 C) 899 D) 905

10. **There are 20 packets with 8 pencils in each. How many pencils are there altogether?**
    A) 152 B) 160 C) 168 D) 28

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 24	2. 200	3. 162
4. 600	5. 72	6. 112
7. 120	8. 64	9. 20
10. 60	11. 200	12. 500
13. 70	14. 900	15. 160

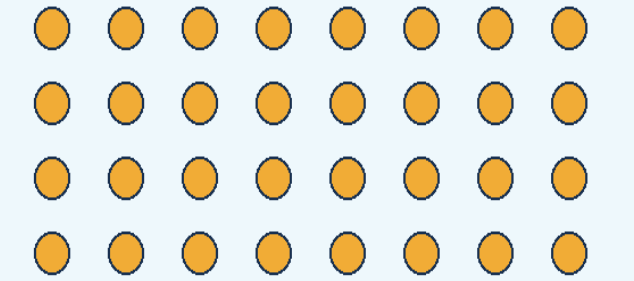
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Multiplying large numbers

LEARNING GOAL Multiplying large numbers

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to multiply multi-digit numbers efficiently.

WORKED EXAMPLE 1

Work out 42 × 2.

Answer: 84

Multiply ones, then tens and hundreds; total 84.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Multiplying large numbers

PLAY · LEARN THE STEPS

1. **Work out 195 × 9.**
   A) 1752 B) 1755 C) 1750 D) 1757
2. **Work out 103 × 8.**
   A) 827 B) 1824 C) 824 D) 820
3. **Work out 269 × 2.**
   A) 518 B) 538 C) 536 D) 588
4. **Work out 233 × 6.**
   A) 1398 B) 1408 C) 1399 D) 1418
5. **Work out 192 × 7.**
   A) 1349 B) 1344 C) 1339 D) 1343

CHALLENGE · DIFFERENT EXAMPLES

1. **There are 200 trays with 5 oranges in each. How many oranges are there altogether?**
   A) 205 B) 995 C) 1005 D) 1000
2. **Work out 237 × 3.**
   A) 708 B) 811 C) 711 D) 731
3. **There are 187 rows with 5 chairs in each. How many chairs are there altogether?**
   A) 930 B) 192 C) 935 D) 940
4. **Work out 195 × 4.**
   A) 680 B) 780 C) 730 D) 790
5. **There are 236 packets with 9 pencils in each. How many pencils are there altogether?**
   A) 2124 B) 245 C) 2133 D) 2115

Multiplying large numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **Work out 145 × 5.**
A) 1725 B) 728 C) 725 D) 726

7. **There are 130 teams with 7 players in each. How many players are there altogether?**
A) 910 B) 137 C) 917 D) 903

8. **Work out 25 × 2.**
A) 46 B) 53 C) 50 D) 30

9. **There are 87 tables with 7 books in each. How many books are there altogether?**
A) 602 B) 616 C) 609 D) 94

10. **Work out 322 × 2.**
A) 644 B) 647 C) 639 D) 645

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 1755	2. 824	3. 538
4. 1398	5. 1344	6. 1000
7. 711	8. 935	9. 780
10. 2124	11. 725	12. 910
13. 50	14. 609	15. 644

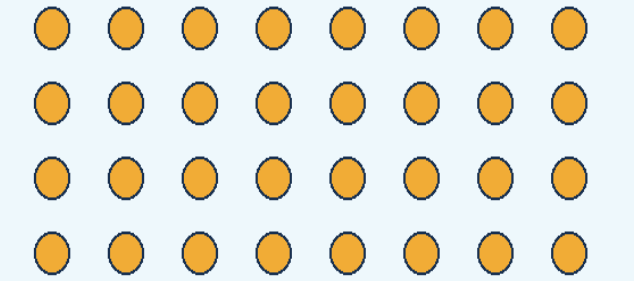
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Division facts

LEARNING GOAL Division facts

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to determine basic division fact up to 81.

WORKED EXAMPLE 1

Find 4 ÷ 2.

Answer: 2

Because 2 × 2 = 4, 4 ÷ 2 = 2.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Division facts

PLAY · LEARN THE STEPS

1. **Find 48 ÷ 6.**
   A) 8 B) 10 C) 7 D) 108
2. **Find 72 ÷ 8.**
   A) 9 B) 59 C) 109 D) 6
3. **Find 63 ÷ 9.**
   A) 7 B) 2 C) 1007 D) 10
4. **Find 18 ÷ 6.**
   A) 7 B) 23 C) 3 D) 2
5. **Find 20 ÷ 5.**
   A) 1004 B) 14 C) 4 D) 0

CHALLENGE · DIFFERENT EXAMPLES

1. **A teacher puts 36 mangoes into sets of 4. How many full sets can she make?**
   A) 4 B) 13 C) 10 D) 9
2. **Find 42 ÷ 7.**
   A) 7 B) 2 C) 6 D) 4
3. **48 pencils are shared equally among 8 groups. How many go to each?**
   A) 40 B) 14 C) 6 D) 8
4. **Find 14 ÷ 7.**
   A) 202 B) 2 C) 0 D) 52
5. **A teacher puts 56 stickers into sets of 8. How many full sets can she make?**
   A) 9 B) 15 C) 7 D) 8

Division facts

CHALLENGE · APPLY WHAT YOU KNOW

6. **Find 4 ÷ 2.**
A) 4 B) 2 C) 0 D) 6

7. **33 books are shared equally among 3 shelves. How many go to each?**
A) 14 B) 30 C) 3 D) 11

8. **A teacher puts 90 plant pots into sets of 6. How many full sets can she make?**
A) 15 B) 6 C) 16 D) 21

9. **Find 28 ÷ 7.**
A) 9 B) 2 C) 4 D) 1004

10. **120 oranges are shared equally among 8 crates. How many go to each?**
A) 15 B) 112 C) 8 D) 23

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 8	2. 9	3. 7
4. 3	5. 4	6. 9
7. 6	8. 6	9. 2
10. 7	11. 2	12. 11
13. 15	14. 4	15. 15

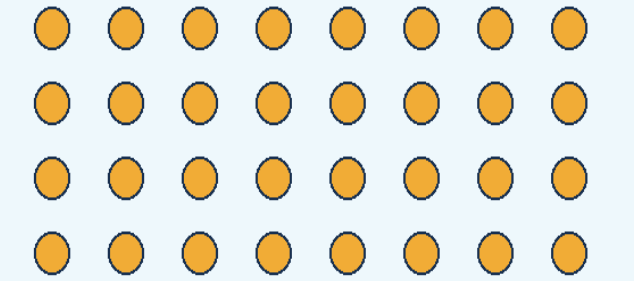
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Dividing by one-digit numbers

LEARNING GOAL Dividing by one-digit numbers

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to divide 2-digit numbers by 1-digit number efficiently.

WORKED EXAMPLE 1

Divide 4 ÷ 2.

Answer: 2

2 × 2 = 4.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Dividing by one-digit numbers

PLAY · LEARN THE STEPS

1. **Divide 36 ÷ 4.**
   A) 7 B) 14 C) 4 D) 9
2. **Divide 16 ÷ 2.**
   A) 12 B) 1008 C) 3 D) 8
3. **Divide 30 ÷ 5.**
   A) 1006 B) 8 C) 1 D) 6
4. **Divide 63 ÷ 9.**
   A) 57 B) 107 C) 11 D) 7
5. **Divide 54 ÷ 6.**
   A) 11 B) 1009 C) 9 D) 109

CHALLENGE · DIFFERENT EXAMPLES

1. **A teacher puts 244 mangoes into sets of 4. How many full sets can she make?**
   A) 61 B) 62 C) 4 D) 65
2. **Divide 90 ÷ 9.**
   A) 7 B) 10 C) 0 D) 15
3. **90 pencils are shared equally among 6 groups. How many go to each?**
   A) 6 B) 15 C) 84 D) 21
4. **Divide 30 ÷ 3.**
   A) 10 B) 60 C) 210 D) 30
5. **A teacher puts 207 stickers into sets of 9. How many full sets can she make?**
   A) 24 B) 9 C) 23 D) 32

Dividing by one-digit numbers

CHALLENGE · APPLY WHAT YOU KNOW

6. **Divide 42 ÷ 6.**
A) 5 B) 11 C) 4 D) 7

7. **558 books are shared equally among 9 shelves. How many go to each?**
A) 549 B) 62 C) 9 D) 71

8. **Divide 6 ÷ 2.**
A) 3 B) 4 C) 8 D) 13

9. **A teacher puts 26 plant pots into sets of 2. How many full sets can she make?**
A) 14 B) 15 C) 13 D) 2

10. **Divide 20 ÷ 2.**
A) 7 B) 10 C) 11 D) 60

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 9	2. 8	3. 6
4. 7	5. 9	6. 61
7. 10	8. 15	9. 10
10. 23	11. 7	12. 62
13. 3	14. 13	15. 10

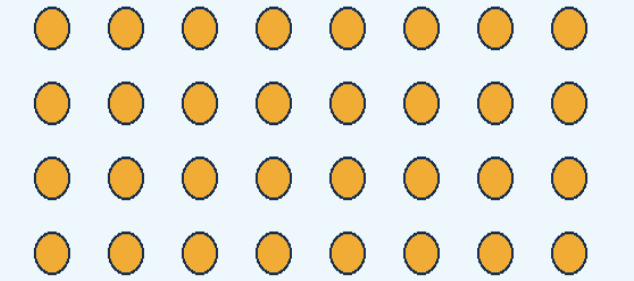
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Two-step word problems

LEARNING GOAL Two-step word problems

MATHS MODEL · NUMBERS AND OPERATIONS



4 rows × 8 columns = 32 objects

LEARN THE IDEA

Learn to solve multi-step word problems involving the four basic operations.

WORKED EXAMPLE 1

A class collected 15 blue and 7 green bottle tops, then shared them into 7 equal groups. How many complete bottle tops per group?

Answer: 3

15+7=22; divide by 7 for 3 complete items (with 1 left over).

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Two-step word problems

PLAY · LEARN THE STEPS

1. **A class collected 21 blue and 6 green bottle tops, then shared them into 4 equal groups. How many complete bottle tops per group?**
   A) 6 B) 1006 C) 206 D) 10
2. **A class collected 47 blue and 7 green bottle tops, then shared them into 8 equal groups. How many complete bottle tops per group?**
   A) 1 B) 6 C) 8 D) 56
3. **A class collected 26 blue and 34 green bottle tops, then shared them into 2 equal groups. How many complete bottle tops per group?**
   A) 34 B) 25 C) 30 D) 80
4. **A class collected 19 blue and 28 green bottle tops, then shared them into 5 equal groups. How many complete bottle tops per group?**
   A) 5 B) 13 C) 14 D) 9
5. **A class collected 22 blue and 32 green bottle tops, then shared them into 7 equal groups. How many complete bottle tops per group?**
   A) 12 B) 7 C) 27 D) 17

CHALLENGE · DIFFERENT EXAMPLES

1. **A class has 21 blue beads and 17 red beads. It makes 5 equal necklaces. How many whole items can each get?**
   A) 6 B) 38 C) 8 D) 7
2. **A class collected 56 blue and 38 green bottle tops, then shared them into 6 equal groups. How many complete bottle tops per group?**
   A) 10 B) 5 C) 15 D) 65
3. **A class has 17 mangoes and 17 oranges. It makes 9 equal baskets. How many whole items can each get?**
   A) 3 B) 2 C) 4 D) 34
4. **A class collected 18 blue and 33 green bottle tops, then shared them into 5 equal groups. How many complete bottle tops per group?**
   A) 1010 B) 8 C) 13 D) 10
5. **A class has 13 pencils and 11 crayons. It makes 8 equal boxes. How many whole items can each get?**
   A) 4 B) 3 C) 2 D) 24

Two-step word problems

CHALLENGE · APPLY WHAT YOU KNOW

6 **A class collected 12 blue and 32 green bottle tops, then shared them into 7 equal groups. How many complete bottle tops per group?**

A) 1006 B) 4 C) 6 D) 56

7 **A class has 35 drums and 23 balls. It makes 4 equal teams. How many whole items can each get?**

A) 58 B) 15 C) 13 D) 14

8 **A class collected 26 blue and 30 green bottle tops, then shared them into 8 equal groups. How many complete bottle tops per group?**

A) 2 B) 7 C) 11 D) 9

9 **A class has 35 blue beads and 14 red beads. It makes 8 equal necklaces. How many whole items can each get?**

A) 7 B) 5 C) 6 D) 49

10 **A class collected 12 blue and 11 green bottle tops, then shared them into 9 equal groups. How many complete bottle tops per group?**

A) 22 B) 12 C) 2 D) 1002

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 6	2. 6	3. 30
4. 9	5. 7	6. 7
7. 15	8. 3	9. 10
10. 3	11. 6	12. 14
13. 7	14. 6	15. 2

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Unit fractions on number lines & Equivalent fractions

LEARNING GOAL Unit fractions on number lines / Equivalent fractions

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to partition the 0 to 1 number line into equal unit fractions and recognise and name equivalent fractions using pictorial representations and number line to determine the lowest common denominator (lcd).

WORKED EXAMPLE 1

A number line from 0 to 1 is divided into 2 equal parts. What fraction is at mark 1 after zero?

Answer: 1/2

Each tick is one 2th, so mark 1 is 1/2.

WORKED EXAMPLE 2

Which fraction is equivalent to 1/2?

Answer: 6/12

Multiply numerator and denominator by 6.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Unit fractions on number lines & Equivalent fractions

PLAY · LEARN THE STEPS

1. **A number line from 0 to 1 is divided into 6 equal parts. What fraction is at mark 4 after zero?**
   A) 4/6 B) 4/8 C) 4/4 D) 4/5
2. **Which fraction is equivalent to 2/3?**
   A) 15/10 B) 10/8 C) 7/15 D) 10/15
3. **A number line from 0 to 1 is divided into 4 equal parts. What fraction is at mark 3 after zero?**
   A) 3/10 B) 3/5 C) 3/12 D) 3/4
4. **Which fraction is equivalent to 6/12?**
   A) 36/72 B) 36/18 C) 12/72 D) 72/36
5. **A number line from 0 to 1 is divided into 8 equal parts. What fraction is at mark 6 after zero?**
   A) 5/8 B) 6/8 C) 6/4 D) 6/6

CHALLENGE · DIFFERENT EXAMPLES

1. **A measuring strip is divided into 8 equal sections. 4 are coloured. What fraction is coloured?**
   A) 5/8 B) 4/9 C) 4/8 D) 8/4
2. **A number line from 0 to 1 is divided into 2 equal parts. What fraction is at mark 1 after zero?**
   A) 1/4 B) 1/10 C) 2/2 D) 1/2
3. **A measuring strip is divided into 6 equal sections. 2 are coloured. What fraction is coloured?**
   A) 6/2 B) 2/6 C) 3/6 D) 2/7
4. **A measuring strip is divided into 4 equal sections. 3 are coloured. What fraction is coloured?**
   A) 4/4 B) 3/5 C) 3/4 D) 4/3
5. **A number line from 0 to 1 is divided into 3 equal parts. What fraction is at mark 2 after zero?**
   A) 2/3 B) 1/3 C) 2/5 D) 2/4

Unit fractions on number lines & Equivalent fractions

CHALLENGE · APPLY WHAT YOU KNOW

6. **A recipe needs 2/5 cup. Which equivalent fraction has denominator 25?**
   A) 7/25 B) 10/5 C) 10/25 D) 2/25

7. **Which fraction is equivalent to 1/4?**
   A) 6/24 B) 24/6 C) 6/10 D) 7/24

8. **A recipe needs 5/6 cup. Which equivalent fraction has denominator 30?**
   A) 5/30 B) 25/6 C) 10/30 D) 25/30

9. **Which fraction is equivalent to 1/3?**
   A) 18/6 B) 7/18 C) 6/9 D) 6/18

10. **A recipe needs 3/5 cup. Which equivalent fraction has denominator 10?**
    A) 5/10 B) 3/10 C) 6/5 D) 6/10

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 4/6	2. 10/15	3. 3/4
4. 36/72	5. 6/8	6. 4/8
7. 1/2	8. 2/6	9. 3/4
10. 2/3	11. 10/25	12. 6/24
13. 25/30	14. 6/18	15. 6/10

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Simplifying fractions & Improper fractions

LEARNING GOAL Simplifying fractions / Improper fractions

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to find the simplest form of given fractions by dividing through by the highest common factor (i.e. by cancelling through by factors) and recognise fractions that are greater than one (i.e. improper fractions), draw and label such fractions with their symbols.

WORKED EXAMPLE 1

Simplify 12/24 to lowest terms.

Answer: 1/2

Divide by 2 and simplify any other common factor.

WORKED EXAMPLE 2

Write the improper fraction 17/7 as a mixed number.

Answer: 2 3/7

17 = 2 × 7 + 3.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Simplifying fractions & Improper fractions

PLAY · LEARN THE STEPS

1. **Simplify 42/84 to lowest terms.**
   A) 6/84 B) 42/12 C) 1/2 D) 7/12
2. **Write the improper fraction 42/9 as a mixed number.**
   A) 3 6/9 B) 4 3/9 C) 5 6/9 D) 4 6/9
3. **Simplify 12/26 to lowest terms.**
   A) 12/13 B) 6/13 C) 7/13 D) 6/26
4. **Write the improper fraction 8/6 as a mixed number.**
   A) 2 2/6 B) 0 2/6 C) 1 4/6 D) 1 2/6
5. **Simplify 18/20 to lowest terms.**
   A) 18/10 B) 9/10 C) 9/20 D) 10/10

CHALLENGE · DIFFERENT EXAMPLES

1. **A shaded diagram shows 18 out of 21 equal squares. What is this fraction in lowest terms?**
   A) 6/21 B) 6/7 C) 18/21 D) 18/7
2. **A shaded diagram shows 18 out of 54 equal squares. What is this fraction in lowest terms?**
   A) 18/54 B) 1/3 C) 3/54 D) 18/9
3. **Simplify 21/30 to lowest terms.**
   A) 21/10 B) 7/30 C) 8/10 D) 7/10
4. **A shaded diagram shows 15 out of 25 equal squares. What is this fraction in lowest terms?**
   A) 15/5 B) 3/5 C) 15/25 D) 3/25
5. **A shaded diagram shows 16 out of 8 equal squares. What is this fraction in lowest terms?**
   A) 16/8 B) 16/4 C) 8/8 D) 2/1

Simplifying fractions & Improper fractions

CHALLENGE · APPLY WHAT YOU KNOW

6. **41 equal pieces are grouped 9 per whole. Write 41/9 as a mixed number.**
A) 4/9 B) 5 5/9 C) 41 5/9 D) 4 5/9

7. **Write the improper fraction 44/9 as a mixed number.**
A) 5 8/9 B) 4 8/9 C) 4 1/9 D) 3 8/9

8. **26 equal pieces are grouped 5 per whole. Write 26/5 as a mixed number.**
A) 6 1/5 B) 26 1/5 C) 5 1/5 D) 5/5

9. **Write the improper fraction 22/5 as a mixed number.**
A) 4 3/5 B) 4 2/5 C) 3 2/5 D) 5 2/5

10. **28 equal pieces are grouped 10 per whole. Write 28/10 as a mixed number.**
A) 2 8/10 B) 3 8/10 C) 2/10 D) 28 8/10

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 1/2	2. 4 6/9	3. 6/13
4. 1 2/6	5. 9/10	6. 6/7
7. 1/3	8. 7/10	9. 3/5
10. 2/1	11. 4 5/9	12. 4 8/9
13. 5 1/5	14. 4 2/5	15. 2 8/10

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing fractions & Everyday fractions

LEARNING GOAL Comparing fractions / Everyday fractions

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to compare and order fractions with like denominators by using pictorial representations and finding equivalent fractions using the lowest common denominator (lcd) and provide examples of where fractions are used.

WORKED EXAMPLE 1

Which fraction is greater: 1/4 or 3/4?

Answer: 3/4

With equal denominators, the larger numerator gives the larger fraction.

WORKED EXAMPLE 2

A family has 15 equal units of a cloth length. How many units make one third?

Answer: 5

A third is 1/3 of 15, which equals 5.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Comparing fractions & Everyday fractions

PLAY · LEARN THE STEPS

1. **Which fraction is greater: 2/4 or 3/4?**
   A) 1/4 B) 3/4 C) 2/4
2. **A family has 30 equal units of a group of mangoes. How many units make one half?**
   A) 16 B) 10 C) 15 D) 65
3. **Which fraction is greater: 1/6 or 3/6?**
   A) 3/6s B) 3/6 C) not 3/6 D) 1/6
4. **A family has 26 equal units of a bucket of water. How many units make one half?**
   A) 17 B) 12 C) 8 D) 13
5. **Which fraction is greater: 4/8 or 7/8?**
   A) 4/8 B) 7/8 C) 1/8

CHALLENGE · DIFFERENT EXAMPLES

1. **Two ribbons are 2/8 m and 7/8 m long. Which is longer?**
   A) 7/8 B) 9/8 C) 2/8 D) 8/7
2. **Two ribbons are 4/6 m and 3/6 m long. Which is longer?**
   A) 6/4 B) 4/6 C) 7/6 D) 3/6
3. **Which fraction is greater: 1/5 or 2/5?**
   A) 2/5 B) 1/5 C) 3/5
4. **Two ribbons are 5/6 m and 2/6 m long. Which is longer?**
   A) 6/5 B) 7/6 C) 5/6 D) 2/6
5. **Which fraction is greater: 3/10 or 5/10?**
   A) 5/10 B) 1/10 C) 3/10

Comparing fractions & Everyday fractions

CHALLENGE · APPLY WHAT YOU KNOW

6. **A class has 15 bottle tops. It gives 2/3 of them to a club. How many are given?**
A) 13 B) 10 C) 5 D) 12

7. **A family has 36 equal units of a loaf. How many units make one third?**
A) 12 B) 15 C) 62 D) 17

8. **A class has 10 bottle tops. It gives 1/2 of them to a club. How many are given?**
A) 7 B) 10 C) 9 D) 5

9. **A family has 32 equal units of a bucket of water. How many units make one quarter?**
A) 28 B) 10 C) 7 D) 8

10. **A class has 30 bottle tops. It gives 4/5 of them to a club. How many are given?**
A) 26 B) 6 C) 24 D) 29

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 3/4	2. 15	3. 3/6
4. 13	5. 7/8	6. 7/8
7. 4/6	8. 2/5	9. 5/6
10. 5/10	11. 10	12. 12
13. 5	14. 8	15. 24

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Tenths and hundredths & Rounding decimals

LEARNING GOAL Tenths and hundredths / Rounding decimals

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to describe and represent decimals (tenths and hundredths) concretely, pictorially, and symbolically and round decimals to the nearest tenth.

WORKED EXAMPLE 1

0 tenths and 0 hundredths make which decimal?

Answer: 0.00

0/10 + 0/100 = 0.00.

WORKED EXAMPLE 2

Round 5.03 to the nearest tenth.

Answer: 5.0

Check the hundredths digit: 5 or more rounds the tenths up.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Tenths and hundredths & Rounding decimals

PLAY · LEARN THE STEPS

1. **0 tenths and 7 hundredths make which decimal?**
   A) 0.70 B) 0.08 C) 0.07 D) 0.7
2. **Round 0.04 to the nearest tenth.**
   A) 0.0 B) 0.1 C) 0.04
3. **9 tenths and 9 hundredths make which decimal?**
   A) 0.99 B) 9.9 C) 0.90
4. **Round 5.11 to the nearest tenth.**
   A) 5.0 B) 5.2 C) 5.11 D) 5.1
5. **1 tenths and 1 hundredths make which decimal?**
   A) 0.12 B) 1.1 C) 0.11

CHALLENGE · DIFFERENT EXAMPLES

1. **A ruler reading has 3 tenths and 8 hundredths of a metre. What decimal length is this?**
   A) 0.38 B) 0.48 C) 3.80 D) 0.37
2. **9 tenths and 3 hundredths make which decimal?**
   A) 0.93 B) 0.94 C) 0.39 D) 9.3
3. **A ruler reading has 3 tenths and 3 hundredths of a metre. What decimal length is this?**
   A) 0.43 B) 0.32 C) 3.30 D) 0.33
4. **2 tenths and 3 hundredths make which decimal?**
   A) 0.24 B) 0.32 C) 0.23 D) 2.3
5. **A ruler reading has 2 tenths and 2 hundredths of a metre. What decimal length is this?**
   A) 0.21 B) 0.22 C) 2.20 D) 0.32

Tenths and hundredths & Rounding decimals

CHALLENGE · APPLY WHAT YOU KNOW

6. **A bottle holds 4.34 litres. Rounded to one decimal place, how much is that?**
   A) 4.2 B) 4.34 C) 4.3 D) 4.4

7. **Round 4.22 to the nearest tenth.**
   A) 4.3 B) 4.22 C) 4.2 D) 4.1

8. **A bottle holds 5.24 litres. Rounded to one decimal place, how much is that?**
   A) 5.24 B) 5.2 C) 5.3 D) 5.1

9. **Round 2.50 to the nearest tenth.**
   A) 2.50 B) 2.5 C) 2.4 D) 2.6

10. **A bottle holds 5.27 litres. Rounded to one decimal place, how much is that?**
    A) 5.27 B) 5.4 C) 5.2 D) 5.3

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 0.07	2. 0.0	3. 0.99
4. 5.1	5. 0.11	6. 0.38
7. 0.93	8. 0.33	9. 0.23
10. 0.22	11. 4.3	12. 4.2
13. 5.2	14. 2.5	15. 5.3

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Adding and subtracting decimals

LEARNING GOAL Adding and subtracting decimals

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to use models to explain the result of addition and subtraction of decimals (up to hundredths).

WORKED EXAMPLE 1

Calculate 0.12 − 0.04.

Answer: 0.08

Line up the decimal points and combine hundredths.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Adding and subtracting decimals

PLAY · LEARN THE STEPS

1. **Calculate 0.40 + 0.14.**
   A) 0.54 B) 0.64 C) 0.44 D) 0.55
2. **Calculate 0.32 − 0.04.**
   A) 0.29 B) 0.18 C) 0.38 D) 0.28
3. **Calculate 0.35 − 0.12.**
   A) 0.24 B) 0.13 C) 0.23 D) 0.33
4. **Calculate 0.49 + 0.10.**
   A) 0.69 B) 0.60 C) 0.49 D) 0.59
5. **Calculate 0.66 + 0.19.**
   A) 0.85 B) 0.75 C) 0.86 D) 0.95

CHALLENGE · DIFFERENT EXAMPLES

1. **Kofi cuts 2.7 m from a 6.0 m rope. What length remains?**
   A) 8.7 B) 3.4 C) 4.3 D) 3.3
2. **Calculate 0.55 − 0.09.**
   A) 0.36 B) 0.56 C) 0.46 D) 0.47
3. **A shopper buys soap for GH₵2.2 and bread for GH₵2.5. What is the total?**
   A) 4.7 B) 5.7 C) 4.8 D) 3.7
4. **Calculate 0.43 + 0.16.**
   A) 0.69 B) 0.60 C) 0.59 D) 0.49
5. **Kofi cuts 2.6 m from a 3.5 m rope. What length remains?**
   A) 0.9 B) 1.9 C) 1.0 D) 6.1

Adding and subtracting decimals

CHALLENGE · APPLY WHAT YOU KNOW

6. **Calculate 0.57 + 0.12.**
A) 0.59 B) 0.69 C) 0.70 D) 0.79

7. **A shopper buys soap for GH¢1.6 and bread for GH¢2.2. What is the total?**
A) 4.8 B) 2.8 C) 3.8 D) 3.9

8. **Calculate 0.18 − 0.07.**
A) 0.12 B) 0.21 C) 0.01 D) 0.11

9. **Kofi cuts 1.8 m from a 3.6 m rope. What length remains?**
A) 1.8 B) 2.8 C) 1.9 D) 5.4

10. **Calculate 0.33 − 0.07.**
A) 0.27 B) 0.36 C) 0.16 D) 0.26

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 0.54	2. 0.28	3. 0.23
4. 0.59	5. 0.85	6. 3.3
7. 0.46	8. 4.7	9. 0.59
10. 0.9	11. 0.69	12. 3.8
13. 0.11	14. 1.8	15. 0.26

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Percent out of 100 & Comparing fractions, decimals and percents

LEARNING GOAL Percent out of 100 / Comparing fractions, decimals and percents

MATHS MODEL · NUMBERS AND OPERATIONS



6/10 = 0.6 = 60%

LEARN THE IDEA

Learn to model or recognise percent (as a fraction related to hundredths) using concrete models, pictorial representations and number line and compare and order a mixture of common, decimal and percent fractions (up to hundredths).

WORKED EXAMPLE 1

Out of 100 equally sized squares, 12 are shaded. What percent is shaded?

Answer: 12%

12 out of 100 means 12%.

WORKED EXAMPLE 2

Which is larger: 89% or 0.01?

Answer: 89%

Write both as hundredths before comparing.

TRY THIS IN REAL LIFE

Use a number chart, ten-stick bundles or paper place-value tokens. Model your answer, then explain why it works.

Percent out of 100 & Comparing fractions, decimals and percents

PLAY · LEARN THE STEPS

1. **Out of 100 equally sized squares, 88 are shaded. What percent is shaded?**
   A) 12% B) 83% C) 93% D) 88%
2. **Which is larger: 13% or 0.60?**
   A) 60% B) 47% C) 73% D) 13%
3. **Out of 100 equally sized squares, 73 are shaded. What percent is shaded?**
   A) 73% B) 78% C) 68% D) 27%
4. **Which is larger: 32% or 0.56?**
   A) 56% B) 32% C) 24% D) 88%
5. **Out of 100 equally sized squares, 80 are shaded. What percent is shaded?**
   A) 85% B) 20% C) 80% D) 75%

CHALLENGE · DIFFERENT EXAMPLES

1. **Out of 100 pupils, 40 prefer reading. What percentage is that?**
   A) 40/10% B) 30% C) 50% D) 40%
2. **Out of 100 equally sized squares, 12 are shaded. What percent is shaded?**
   A) 7% B) 12% C) 17% D) 88%
3. **Out of 100 pupils, 20 prefer reading. What percentage is that?**
   A) 10% B) 20% C) 30% D) 20/10%
4. **Out of 100 equally sized squares, 54 are shaded. What percent is shaded?**
   A) 59% B) 46% C) 54% D) 49%
5. **Out of 100 equally sized squares, 51 are shaded. What percent is shaded?**
   A) 46% B) 51% C) 49% D) 56%

Percent out of 100 & Comparing fractions, decimals and percents

CHALLENGE · APPLY WHAT YOU KNOW

6. **A shop discount is 75%. What decimal expresses the same proportion?**
   A) 0.8 B) 75 C) 7.5 D) 0.75

7. **Which is larger: 26% or 0.06?**
   A) 6% B) 32% C) 20% D) 26%

8. **A shop discount is 40%. What decimal expresses the same proportion?**
   A) 40 B) 0.4 C) 4 D) 0.45

9. **Which is larger: 36% or 0.81?**
   A) 117% B) 45% C) 36% D) 81%

10. **A shop discount is 50%. What decimal expresses the same proportion?**
    A) 0.55 B) 50 C) 5 D) 0.5

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 88%	2. 60%	3. 73%
4. 56%	5. 80%	6. 40%
7. 12%	8. 20%	9. 54%
10. 51%	11. 0.75	12. 26%
13. 0.4	14. 81%	15. 0.5

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Recognising number patterns & Finding missing terms

LEARNING GOAL Recognising number patterns / Finding missing terms

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to describe the pattern found in a given table or chart and determine the missing element(s) in a given table or chart.

WORKED EXAMPLE 1

Which rule describes the pattern 6, 9, 12, ...?

Answer: Add 3

Each next term increases by 3.

WORKED EXAMPLE 2

Fill the missing number: 3, 12, ___, 30.

Answer: 21

The rule is add 9.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Recognising number patterns & Finding missing terms

PLAY · LEARN THE STEPS

1. **Which rule describes the pattern 17, 30, 43, ...?**
   A) Multiply by 13 B) Add 13 C) Add 14 D) Subtract 13
2. **Fill the missing number: 15, 17, ___, 21.**
   A) 22 B) 19 C) 17 D) 29
3. **Which rule describes the pattern 37, 50, 63, ...?**
   A) Add 13 B) Add 14 C) Subtract 13 D) Multiply by 13
4. **Fill the missing number: 24, 40, ___, 72.**
   A) 56 B) 36 C) 256 D) 6
5. **Which rule describes the pattern 16, 27, 38, ...?**
   A) Add 11 B) Subtract 11 C) Multiply by 11 D) Add 12

CHALLENGE · DIFFERENT EXAMPLES

1. **A walking counter records 16, 28, 40 steps. What is the change each time?**
   A) +12 B) +13 C) −12 D) +24
2. **Which rule describes the pattern 24, 32, 40, ...?**
   A) Subtract 8 B) Add 9 C) Add 8 D) Multiply by 8
3. **A walking counter records 30, 39, 48 steps. What is the change each time?**
   A) +18 B) +10 C) −9 D) +9
4. **Which rule describes the pattern 22, 34, 46, ...?**
   A) Add 12 B) Subtract 12 C) Add 13 D) Multiply by 12
5. **A walking counter records 19, 23, 27 steps. What is the change each time?**
   A) +5 B) +8 C) +4 D) −4

Recognising number patterns & Finding missing terms

CHALLENGE · APPLY WHAT YOU KNOW

6. **A growing drawing has 34, 46, __, 70 dots. How many dots are missing?**
   A) 46 B) 70 C) 58 D) 59

7. **Fill the missing number: 3, 12, ___, 30.**
   A) 18 B) 21 C) 25 D) 23

8. **A growing drawing has 16, 24, __, 40 dots. How many dots are missing?**
   A) 24 B) 40 C) 33 D) 32

9. **A growing drawing has 12, 19, __, 33 dots. How many dots are missing?**
   A) 33 B) 26 C) 27 D) 19

10. **A growing drawing has 5, 13, __, 29 dots. How many dots are missing?**
    A) 21 B) 22 C) 13 D) 29

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. Add 13	2. 19	3. Add 13
4. 56	5. Add 11	6. +12
7. Add 8	8. +9	9. Add 12
10. +4	11. 58	12. 21
13. 32	14. 26	15. 21

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Finding pattern errors

LEARNING GOAL Finding pattern errors

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to identify the error(s) in a given table or chart.

WORKED EXAMPLE 1

A table should add 3 each time: 6, 9, 15, 15. What should replace 15?

Answer: 12

The correct third entry is 12.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Finding pattern errors

PLAY · LEARN THE STEPS

1. **A table should add 5 each time: 25, 30, 36, 40. What should replace 36?**
   A) 30 B) 55 C) 35 D) 135
2. **A table should add 9 each time: 19, 28, 38, 46. What should replace 38?**
   A) 35 B) 37 C) 57 D) 137
3. **A table should add 7 each time: 36, 43, 51, 57. What should replace 51?**
   A) 52 B) 50 C) 54 D) 48
4. **A table should add 6 each time: 33, 39, 49, 51. What should replace 49?**
   A) 45 B) 43 C) 25 D) 50
5. **A table should add 16 each time: 40, 56, 74, 88. What should replace 74?**
   A) 72 B) 69 C) 71 D) 172

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner says 33, 40, 50, 54 goes up by 7. Which listed term is incorrect?**
   A) 54 B) 33 C) 50 D) 40
2. **A table should add 5 each time: 30, 35, 43, 45. What should replace 43?**
   A) 40 B) 60 C) 1040 D) 44
3. **This chart should increase by 11: 4, 15, 26, 40, 48. Which number must replace 40?**
   A) 38 B) 40 C) 37 D) 48
4. **A table should add 6 each time: 30, 36, 47, 48. What should replace 47?**
   A) 42 B) 47 C) 43 D) 37
5. **A learner says 17, 23, 32, 35 goes up by 6. Which listed term is incorrect?**
   A) 17 B) 23 C) 35 D) 32

Finding pattern errors

CHALLENGE · APPLY WHAT YOU KNOW

6. **A table should add 3 each time: 5, 8, 16, 14. What should replace 16?**
   A) 1 B) 11 C) 21 D) 7

7. **This chart should increase by 13: 25, 38, 51, 69, 77. Which number must replace 69?**
   A) 64 B) 77 C) 65 D) 69

8. **A table should add 6 each time: 32, 38, 47, 50. What should replace 47?**
   A) 44 B) 40 C) 46 D) 43

9. **A learner says 34, 42, 53, 58 goes up by 8. Which listed term is incorrect?**
   A) 42 B) 34 C) 58 D) 53

10. **A table should add 2 each time: 21, 23, 30, 27. What should replace 30?**
    A) 29 B) 21 C) 27 D) 25

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 35	2. 37	3. 50
4. 45	5. 72	6. 50
7. 40	8. 37	9. 42
10. 32	11. 11	12. 64
13. 44	14. 53	15. 25

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Making patterns with objects & Building tables from patterns

LEARNING GOAL Making patterns with objects / Building tables from patterns

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to create a concrete representation of a given pattern displayed in a table or chart and create a table or chart from a given concrete representation of a pattern.

WORKED EXAMPLE 1

A row has 6 dots, the next has 9, and the third has 12. How many in row four?

Answer: 15

Add 3 dots each new row.

WORKED EXAMPLE 2

The pairs in a chart are 1→3, 2→12, 3→21. What number pairs with 5?

Answer: 39

Extend the same table rule.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Making patterns with objects & Building tables from patterns

PLAY · LEARN THE STEPS

1. **A row has 4 dots, the next has 13, and the third has 22. How many in row four?**
   A) 34 B) 31 C) 33 D) 35
2. **The pairs in a chart are 1→9, 2→22, 3→35. What number pairs with 5?**
   A) 63 B) 261 C) 65 D) 61
3. **A row has 11 dots, the next has 23, and the third has 35. How many in row four?**
   A) 27 B) 47 C) 42 D) 44
4. **The pairs in a chart are 1→29, 2→35, 3→41. What number pairs with 5?**
   A) 53 B) 43 C) 3 D) 103
5. **A row has 38 dots, the next has 41, and the third has 44. How many in row four?**
   A) 45 B) 47 C) 42 D) 1047

CHALLENGE · DIFFERENT EXAMPLES

1. **A line of bead rows grows from 34 to 39 to 44. How many beads should row five have?**
   A) 49 B) 54 C) 59 D) 44
2. **A row has 12 dots, the next has 18, and the third has 24. How many in row four?**
   A) 230 B) 30 C) 34 D) 35
3. **A line of bead rows grows from 7 to 14 to 21. How many beads should row five have?**
   A) 35 B) 28 C) 42 D) 21
4. **A row has 32 dots, the next has 45, and the third has 58. How many in row four?**
   A) 21 B) 71 C) 70 D) 66
5. **A line of bead rows grows from 33 to 41 to 49. How many beads should row five have?**
   A) 73 B) 65 C) 49 D) 57

Making patterns with objects & Building tables from patterns

CHALLENGE · APPLY WHAT YOU KNOW

6. **A table pairs input 1→49, 2→64, 3→79. What is the output for input 5?**
   A) 79 B) 94 C) 124 D) 109

7. **The pairs in a chart are 1→29, 2→33, 3→37. What number pairs with 5?**
   A) 35 B) 25 C) 45 D) 245

8. **A table pairs input 1→27, 2→42, 3→57. What is the output for input 5?**
   A) 102 B) 57 C) 72 D) 87

9. **A table pairs input 1→42, 2→51, 3→60. What is the output for input 5?**
   A) 69 B) 87 C) 60 D) 78

10. **The pairs in a chart are 1→28, 2→47, 3→66. What number pairs with 5?**
    A) 102 B) 204 C) 104 D) 108

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 31	2. 61	3. 47
4. 53	5. 47	6. 54
7. 30	8. 35	9. 71
10. 65	11. 109	12. 45
13. 87	14. 78	15. 104

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Turning stories into tables & Using tables to solve problems

LEARNING GOAL Turning stories into tables / Using tables to solve problems

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to translate information from a problem into a table or chart and identify and extend the patterns in a table or chart to solve a given problem.

WORKED EXAMPLE 1

A village saves 6 cedis on Monday and increases savings by 3 cedis each day. What is Thursday's saving?

Answer: 15

Write a daily table and add the same step.

WORKED EXAMPLE 2

A table follows 3, 12, 21, 30, ___. What is its next entry?

Answer: 39

Increase by 9.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Turning stories into tables & Using tables to solve problems

PLAY · LEARN THE STEPS

1. **A village saves 7 cedis on Monday and increases savings by 16 cedis each day. What is Thursday’s saving?**
   A) 50 B) 105 C) 75 D) 55
2. **A table follows 39, 50, 61, 72, ___. What is its next entry?**
   A) 93 B) 33 C) 83 D) 1083
3. **A village saves 15 cedis on Monday and increases savings by 19 cedis each day. What is Thursday’s saving?**
   A) 272 B) 52 C) 72 D) 92
4. **A table follows 31, 34, 37, 40, ___. What is its next entry?**
   A) 47 B) 48 C) 43 D) 39
5. **A village saves 19 cedis on Monday and increases savings by 9 cedis each day. What is Thursday’s saving?**
   A) 47 B) 50 C) 46 D) 66

CHALLENGE · DIFFERENT EXAMPLES

1. **A reading club borrows 35 books on Monday, then 11 more each day. What is Thursday’s total?**
   A) 46 B) 57 C) 79 D) 68
2. **A village saves 35 cedis on Monday and increases savings by 7 cedis each day. What is Thursday’s saving?**
   A) 56 B) 156 C) 61 D) 55
3. **A reading club borrows 18 books on Monday, then 12 more each day. What is Thursday’s total?**
   A) 66 B) 42 C) 30 D) 54
4. **A village saves 30 cedis on Monday and increases savings by 15 cedis each day. What is Thursday’s saving?**
   A) 85 B) 73 C) 78 D) 75
5. **A reading club borrows 3 books on Monday, then 3 more each day. What is Thursday’s total?**
   A) 12 B) 9 C) 6 D) 15

Turning stories into tables & Using tables to solve problems

CHALLENGE · APPLY WHAT YOU KNOW

6 **A savings table is 21, 29, 37, 45. How much more is the fourth entry than the first?**

A) 24 B) 16 C) 8 D) 32

7 **A savings table is 30, 40, 50, 60. How much more is the fourth entry than the first?**

A) 10 B) 20 C) 40 D) 30

8 **A table follows 23, 29, 35, 41, ___. What is its next entry?**

A) 147 B) 47 C) 1047 D) 48

9 **A savings table is 7, 11, 15, 19. How much more is the fourth entry than the first?**

A) 8 B) 16 C) 12 D) 4

10 **A table follows 10, 15, 20, 25, ___. What is its next entry?**

A) 20 B) 25 C) 33 D) 30

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 55	2. 83	3. 72
4. 43	5. 46	6. 68
7. 56	8. 54	9. 75
10. 12	11. 24	12. 30
13. 47	14. 12	15. 30

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Equations from stories & Pictures and missing numbers

LEARNING GOAL Equations from stories / Pictures and missing numbers

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to write a given problem as an equation in which a symbol is used to represent an unknown number and express a given pictorial or concrete representation of an equation in symbolic form.

WORKED EXAMPLE 1

Kofi had some marbles. After receiving 5 more, he has 11. Which equation represents the unknown starting amount?

Answer: n + 5 = 11

Let n be the marbles at the start.

WORKED EXAMPLE 2

The diagram shows an unknown box plus 3 blocks equals 54 blocks. What is in the box?

Answer: 51

54 − 3 = 51.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Equations from stories & Pictures and missing numbers

PLAY · LEARN THE STEPS

1. **Kofi had some marbles. After receiving 33 more, he has 58. Which equation represents the unknown starting amount?**
   A) n + 33 = 58 B) 58 − n = 58 C) n + 58 = 33 D) n − 33 = 58
2. **The diagram shows an unknown box plus 15 blocks equals 49 blocks. What is in the box?**
   A) 35 B) 14 C) 24 D) 34
3. **Kofi had some marbles. After receiving 23 more, he has 40. Which equation represents the unknown starting amount?**
   A) n − 23 = 40 B) n + 40 = 23 C) 40 − n = 40 D) n + 23 = 40
4. **The diagram shows an unknown box plus 6 blocks equals 12 blocks. What is in the box?**
   A) 1006 B) 4 C) 11 D) 6
5. **Kofi had some marbles. After receiving 24 more, he has 55. Which equation represents the unknown starting amount?**
   A) n − 24 = 55 B) n + 55 = 24 C) n + 24 = 55 D) 55 − n = 55

CHALLENGE · DIFFERENT EXAMPLES

1. **Mansa had an unknown number of books; she receives 11 more and now owns 27. Which equation models her original books n?**
   A) 27 + 11 = n B) n + 11 = 27 C) n = 27+11 D) n − 11 = 27
2. **Kofi had some marbles. After receiving 34 more, he has 60. Which equation represents the unknown starting amount?**
   A) 60 − n = 60 B) n + 34 = 60 C) n − 34 = 60 D) n + 60 = 34
3. **Mansa had an unknown number of books; she receives 3 more and now owns 17. Which equation models her original books n?**
   A) n − 3 = 17 B) n = 17+3 C) 17 + 3 = n D) n + 3 = 17
4. **Kofi had some marbles. After receiving 23 more, he has 76. Which equation represents the unknown starting amount?**
   A) n + 23 = 76 B) 76 − n = 76 C) n − 23 = 76 D) n + 76 = 23
5. **Mansa had an unknown number of books; she receives 6 more and now owns 19. Which equation models her original books n?**
   A) n = 19+6 B) 19 + 6 = n C) n − 6 = 19 D) n + 6 = 19

Equations from stories & Pictures and missing numbers

CHALLENGE · APPLY WHAT YOU KNOW

6 **A balance drawing shows n blocks on one side and 6+12 blocks on the other. What must n equal?**

A) 12 B) 18 C) 19 D) 6

7 **The diagram shows an unknown box plus 22 blocks equals 27 blocks. What is in the box?**

A) 8 B) 5 C) 10 D) 25

8 **A balance drawing shows n blocks on one side and 16+15 blocks on the other. What must n equal?**

A) 15 B) 32 C) 31 D) 16

9 **The diagram shows an unknown box plus 14 blocks equals 49 blocks. What is in the box?**

A) 39 B) 36 C) 35 D) 33

10 **A balance drawing shows n blocks on one side and 10+4 blocks on the other. What must n equal?**

A) 15 B) 14 C) 10 D) 4

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. n + 33 = 58	2. 34	3. n + 23 = 40
4. 6	5. n + 24 = 55	6. n + 11 = 27
7. n + 34 = 60	8. n + 3 = 17	9. n + 23 = 76
10. n + 6 = 19	11. 18	12. 5
13. 31	14. 35	15. 14

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Model one-step equations & Explain unknowns

LEARNING GOAL Model one-step equations / Explain unknowns

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to solve a given one-step equation using manipulatives and describe orally, the meaning of a given one-step equation with one unknown.

WORKED EXAMPLE 1

Solve n − 5 = 1.

Answer: 6

Undo subtraction: n=6.

WORKED EXAMPLE 2

The equation is n + 3 = 54. In words, what does n represent?

Answer: the starting number (51)

The unknown n is 51.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Model one-step equations & Explain unknowns

PLAY · LEARN THE STEPS

1. **Solve n + 8 = 33.**
   A) 125 B) 29 C) 20 D) 25
2. **The equation is n − 11 = 31. In words, what does n represent?**
   A) the change (11) B) the result (31) C) the starting number (42) D) a number larger than 62
3. **Solve n − 13 = 5.**
   A) 15 B) 118 C) 22 D) 18
4. **The equation is n + 10 = 31. In words, what does n represent?**
   A) the result (31) B) the starting number (21) C) a number larger than 61 D) the change (10)
5. **Solve n + 15 = 20.**
   A) 55 B) 25 C) 5 D) 1

CHALLENGE · DIFFERENT EXAMPLES

1. **If n + 11 = 36, how many counters represent n?**
   A) 25 B) 11 C) 36 D) 26
2. **Solve n − 33 = -29.**
   A) 104 B) 3 C) 4 D) 1004
3. **If n + 4 = 11, how many counters represent n?**
   A) 11 B) 7 C) 4 D) 8
4. **If n + 5 = 15, how many counters represent n?**
   A) 11 B) 10 C) 15 D) 5
5. **Solve n + 3 = 38.**
   A) 35 B) 25 C) 39 D) 40

Model one-step equations & Explain unknowns

CHALLENGE · APPLY WHAT YOU KNOW

6. **A puzzle says n − 12 = 9. Which number is missing?**
A) 21 B) 12 C) 9 D) 23

7. **The equation is n − 10 = 24. In words, what does n represent?**
A) the starting number (34) B) the change (10) C) the result (24) D) a number larger than 54

8. **A puzzle says n − 14 = 25. Which number is missing?**
A) 41 B) 25 C) 39 D) 14

9. **The equation is n + 26 = 58. In words, what does n represent?**
A) a number larger than 104 B) the change (26) C) the result (58) D) the starting number (32)

10. **A puzzle says n − 10 = 26. Which number is missing?**
A) 26 B) 38 C) 36 D) 10

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 25	2. the starting number (42)	3. 18
4. the starting number (21)	5. 5	6. 25
7. 4	8. 7	9. 10
10. 35	11. 21	12. the starting number (34)
13. 39	14. the starting number (32)	15. 36

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Unknowns on either side & Guess and check

LEARNING GOAL Unknowns on either side / Guess and check

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to solve a given equation when the unknown is on the left or right side of the equation and solve a given one-step equation using “guess and check”.

WORKED EXAMPLE 1

Solve $n - 5 = 1$.

Answer: 6

Undo subtraction: n=6.

WORKED EXAMPLE 2

Guess and check: what number makes $n + 3 = 54$ true?

Answer: 51

Try 51: 51+3=54.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Unknowns on either side & Guess and check

PLAY · LEARN THE STEPS

1. **Solve n − 16 = 19.**
   A) 36 B) 15 C) 85 D) 35
2. **Guess and check: what number makes n + 14 = 17 true?**
   A) 5 B) 3 C) 203 D) 2
3. **Solve 78 = n + 25.**
   A) 49 B) 58 C) 73 D) 53
4. **Guess and check: what number makes n + 20 = 27 true?**
   A) 5 B) 7 C) 4 D) 57
5. **Solve n + 6 = 38.**
   A) 22 B) 32 C) 36 D) 232

CHALLENGE · DIFFERENT EXAMPLES

1. **An unknown number minus 15 equals 13. Which guess is correct?**
   A) 30 B) 28 C) 13 D) 15
2. **An unknown number minus 11 equals 29. Which guess is correct?**
   A) 11 B) 29 C) 42 D) 40
3. **An unknown number minus 9 equals 9. Which guess is correct?**
   A) 23 B) 9 C) 20 D) 18
4. **Solve 0 = n − 7.**
   A) 207 B) 7 C) 9 D) 2
5. **An unknown number minus 13 equals 9. Which guess is correct?**
   A) 24 B) 9 C) 22 D) 13

Unknowns on either side & Guess and check

CHALLENGE · APPLY WHAT YOU KNOW

6. **Kofi checks n + 13 = 29. What value makes his check correct?**
A) 29 B) 18 C) 13 D) 16

7. **Guess and check: what number makes n + 27 = 56 true?**
A) 49 B) 26 C) 30 D) 29

8. **Kofi checks n + 10 = 18. What value makes his check correct?**
A) 10 B) 8 C) 18 D) 13

9. **Guess and check: what number makes n + 5 = 51 true?**
A) 47 B) 51 C) 49 D) 46

10. **Kofi checks n + 14 = 33. What value makes his check correct?**
A) 33 B) 21 C) 14 D) 19

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 35	2. 3	3. 53
4. 7	5. 32	6. 28
7. 40	8. 18	9. 7
10. 22	11. 16	12. 29
13. 8	14. 46	15. 19

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Story equations & Part-part-whole equations

LEARNING GOAL Story equations / Part-part-whole equations

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to identify the unknown in a problem, represent the problem with an equation, and solve the problem concretely, pictorially or symbolically and represent and solve a given addition or subtraction problem involving a “part- part-whole” or comparison context, using a symbol to represent the unknown.

WORKED EXAMPLE 1

11 pencils are in two boxes. One box has 5. Solve n + 5 = 11.

Answer: 6

Subtract the known part from the whole.

WORKED EXAMPLE 2

Two parts are 3 and n; the whole is 54. What is n?

Answer: 51

54 − 3 = 51.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Story equations & Part-part-whole equations

PLAY · LEARN THE STEPS

1. **30 pencils are in two boxes. One box has 8. Solve n + 8 = 30.**
   A) 24 B) 1022 C) 72 D) 22
2. **Two parts are 28 and n; the whole is 45. What is n?**
   A) 217 B) 19 C) 1017 D) 17
3. **77 pencils are in two boxes. One box has 23. Solve n + 23 = 77.**
   A) 64 B) 58 C) 54 D) 55
4. **Two parts are 24 and n; the whole is 72. What is n?**
   A) 28 B) 50 C) 51 D) 48
5. **40 pencils are in two boxes. One box has 30. Solve n + 30 = 40.**
   A) 110 B) 9 C) 12 D) 10

CHALLENGE · DIFFERENT EXAMPLES

1. **Two friends have 33 balls altogether. One has 6. Write the missing amount.**
   A) 27 B) 6 C) 33 D) 39
2. **79 pencils are in two boxes. One box has 26. Solve n + 26 = 79.**
   A) 53 B) 51 C) 33 D) 253
3. **Two friends have 26 balls altogether. One has 12. Write the missing amount.**
   A) 12 B) 14 C) 38 D) 26
4. **48 pencils are in two boxes. One box has 32. Solve n + 32 = 48.**
   A) 14 B) 15 C) 16 D) 66
5. **Two friends have 33 balls altogether. One has 5. Write the missing amount.**
   A) 5 B) 33 C) 28 D) 38

Story equations & Part-part-whole equations

CHALLENGE · APPLY WHAT YOU KNOW

6. **Two parts total 19. One part is 14. How much is the second part?**
   A) 14 B) 7 C) 5 D) 19

7. **Two parts are 4 and n; the whole is 36. What is n?**
   A) 30 B) 32 C) 33 D) 82

8. **Two parts total 32. One part is 23. How much is the second part?**
   A) 23 B) 11 C) 32 D) 9

9. **Two parts are 33 and n; the whole is 78. What is n?**
   A) 48 B) 40 C) 45 D) 50

10. **Two parts total 33. One part is 24. How much is the second part?**
    A) 9 B) 24 C) 11 D) 33

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 22	2. 17	3. 54
4. 48	5. 10	6. 27
7. 53	8. 14	9. 16
10. 28	11. 5	12. 32
13. 9	14. 45	15. 9

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Invent a word problem

LEARNING GOAL Invent a word problem

MATHS MODEL · ALGEBRA AND PATTERNS



Follow the rule. Explain the missing value.

LEARN THE IDEA

Learn to create a problem for a given equation with one unknown.

WORKED EXAMPLE 1

Which word problem fits n + 5 = 11?

Answer: Start with n stickers, get 5 more, finish with 11

The plus sign shows an increase.

TRY THIS IN REAL LIFE

Build a short pattern using bottle tops and draw a table. Ask a partner to explain the rule or solve the missing-number equation.

Invent a word problem

PLAY · LEARN THE STEPS

1. **Which word problem fits n + 8 = 57?**
   A) Multiply n by 8 to get 57 B) Start with n stickers, get 8 more, finish with 57 C) Start with n, lose 8, finish with 57 D) Start with 57, get 8, finish with n
2. **Which word problem fits n + 31 = 57?**
   A) Start with n, lose 31, finish with 57 B) Start with n stickers, get 31 more, finish with 57 C) Multiply n by 31 to get 57 D) Start with 57, get 31, finish with n
3. **Which word problem fits n + 21 = 69?**
   A) Multiply n by 21 to get 69 B) Start with n stickers, get 21 more, finish with 69 C) Start with n, lose 21, finish with 69 D) Start with 69, get 21, finish with n
4. **Which word problem fits n + 20 = 56?**
   A) Start with n, lose 20, finish with 56 B) Multiply n by 20 to get 56 C) Start with n stickers, get 20 more, finish with 56 D) Start with 56, get 20, finish with n
5. **Which word problem fits n + 7 = 51?**
   A) Multiply n by 7 to get 51 B) Start with n stickers, get 7 more, finish with 51 C) Start with n, lose 7, finish with 51 D) Start with 51, get 7, finish with n

CHALLENGE · DIFFERENT EXAMPLES

1. **Which equation would solve “An unknown number of beads, plus 11 more, makes 18”?**
   A) 18 + n = 11 B) n × 11 = 18 C) n − 11 = 18 D) n + 11 = 18
2. **Which word problem fits n + 28 = 54?**
   A) Start with n stickers, get 28 more, finish with 54 B) Multiply n by 28 to get 54 C) Start with n, lose 28, finish with 54 D) Start with 54, get 28, finish with n
3. **Which story represents n + 14 = 31?**
   A) An unknown number minus 14 equals 31 B) An unknown number is divided by 14 to make 31 C) 31 items are removed from 14 items D) An unknown number plus 14 equals 31
4. **Which word problem fits n + 24 = 43?**
   A) Start with n, lose 24, finish with 43 B) Start with n stickers, get 24 more, finish with 43 C) Multiply n by 24 to get 43 D) Start with 43, get 24, finish with n
5. **Which equation would solve “An unknown number of beads, plus 4 more, makes 25”?**
   A) n × 4 = 25 B) n + 4 = 25 C) n − 4 = 25 D) 25 + n = 4

Invent a word problem

CHALLENGE · APPLY WHAT YOU KNOW

6 **Which story represents n + 14 = 28?**

A) An unknown number minus 14 equals 28 B) An unknown number plus 14 equals 28 C) An unknown number is divided by 14 to make 28 D) 28 items are removed from 14 items

7 **Which word problem fits n + 23 = 67?**

A) Start with n, lose 23, finish with 67 B) Start with n stickers, get 23 more, finish with 67 C) Multiply n by 23 to get 67 D) Start with 67, get 23, finish with n

8 **Which equation would solve “An unknown number of beads, plus 13 more, makes 41”?**

A) n + 13 = 41 B) n − 13 = 41 C) n × 13 = 41 D) 41 + n = 13

9 **Which word problem fits n + 21 = 53?**

A) Start with n, lose 21, finish with 53 B) Multiply n by 21 to get 53 C) Start with 53, get 21, finish with n D) Start with n stickers, get 21 more, finish with 53

10 **Which story represents n + 13 = 30?**

A) An unknown number is divided by 13 to make 30 B) An unknown number plus 13 equals 30 C) 30 items are removed from 13 items D) An unknown number minus 13 equals 30

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. Start with n stickers, get 8 more, finish with 57
2. Start with n stickers, get 31 more, finish with 57
3. Start with n stickers, get 21 more, finish with 69
4. Start with n stickers, get 20 more, finish with 56
5. Start with n stickers, get 7 more, finish with 51
6. n + 11 = 18
7. Start with n stickers, get 28 more, finish with 54
8. An unknown number plus 14 equals 31
9. Start with n stickers, get 24 more, finish with 43
10. n + 4 = 25
11. An unknown number plus 14 equals 28
12. Start with n stickers, get 23 more, finish with 67
13. n + 13 = 41
14. Start with n stickers, get 21 more, finish with 53
15. An unknown number plus 13 equals 30

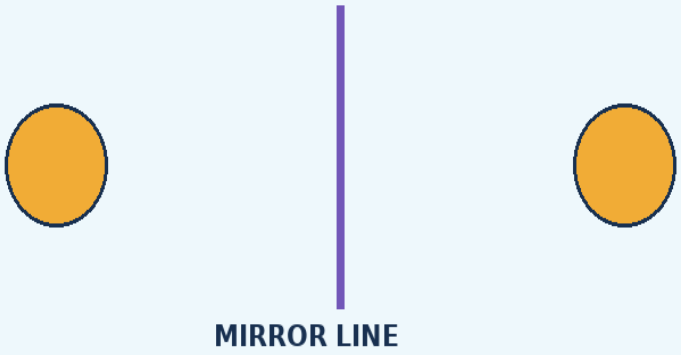
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Draw mirror images & Lines of symmetry

LEARNING GOAL Draw mirror images / Lines of symmetry

MATHS MODEL · GEOMETRY AND MEASUREMENT



LEARN THE IDEA

Learn to complete drawings of shapes to make them symmetrical and identify the lines of symmetry of regular and irregular 2d shapes (triangles and quadrilateral).

WORKED EXAMPLE 1

A point is 2 squares right of a vertical mirror line. Where must its reflection be drawn?

Answer: 2 squares left

A reflection stays the same distance across the mirror line.

WORKED EXAMPLE 2

How many lines of symmetry does a regular decagon have?

Answer: 10

A symmetry line divides a figure into matching halves.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Draw mirror images & Lines of symmetry

PLAY · LEARN THE STEPS

1. **A point is 16 squares below a horizontal mirror line. Where must its reflection be drawn?**
   A) 17 squares above B) 15 squares below C) 16 squares below D) 16 squares above
2. **A regular 5-sided polygon has all equal sides and angles. How many symmetry lines does it have?**
   A) 10 B) 4 C) 5 D) 6
3. **A point is 6 squares right of a vertical mirror line. Where must its reflection be drawn?**
   A) 5 squares right B) 6 squares left C) 7 squares left D) 6 squares right
4. **How many lines of symmetry does a rectangle have?**
   A) 2 B) 5 C) 8 D) 1
5. **A point is 12 squares left of a vertical mirror line. Where must its reflection be drawn?**
   A) 13 squares right B) 11 squares left C) 12 squares left D) 12 squares right

CHALLENGE · DIFFERENT EXAMPLES

1. **A dot is 8 squares left of a vertical mirror. Where should its reflected dot be?**
   A) 9 squares left B) 10 squares right C) 8 squares right D) 8 squares left
2. **A point is 5 squares left of a vertical mirror line. Where must its reflection be drawn?**
   A) 4 squares left B) 5 squares left C) 6 squares right D) 5 squares right
3. **A dot is 5 squares left of a vertical mirror. Where should its reflected dot be?**
   A) 5 squares right B) 5 squares left C) 7 squares right D) 6 squares left
4. **A point is 17 squares left of a vertical mirror line. Where must its reflection be drawn?**
   A) 18 squares right B) 17 squares right C) 16 squares left D) 17 squares left
5. **A dot is 3 squares left of a vertical mirror. Where should its reflected dot be?**
   A) 4 squares left B) 3 squares left C) 5 squares right D) 3 squares right

Draw mirror images & Lines of symmetry

CHALLENGE · APPLY WHAT YOU KNOW

6 **How many lines of symmetry does a isosceles triangle have?**

A) 4 B) 6 C) 1 D) 0

7 **A regular 3-sided polygon has all equal sides and angles. How many symmetry lines does it have?**

A) 3 B) 6 C) 2 D) 4

8 **How many lines of symmetry does a square have?**

A) 3 B) 6 C) 4 D) 5

9 **How many lines of symmetry does a regular decagon have?**

A) 6 B) 5 C) 4 D) 10

10 **How many lines of symmetry does a non-rectangular parallelogram have?**

A) 0 B) 3 C) 10 D) 8

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 16 squares above	2. 5	3. 6 squares left
4. 2	5. 12 squares right	6. 8 squares right
7. 5 squares right	8. 5 squares right	9. 17 squares right
10. 3 squares right	11. 1	12. 3
13. 4	14. 10	15. 0

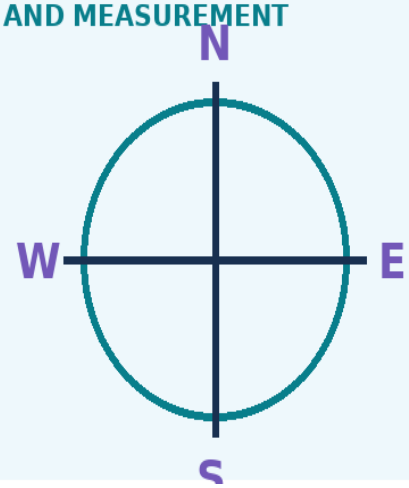
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Compass directions

LEARNING GOAL Compass directions

MATHS MODEL · GEOMETRY AND MEASUREMENT



LEARN THE IDEA

Learn to tell the position and motion of objects in space using the cardinal points north, south, east and west.

WORKED EXAMPLE 1

A pupil walks 3 blocks north. Which compass direction is the pupil moving?

Answer: north

Use a compass rose labelled N, S, E and W.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Compass directions

PLAY · LEARN THE STEPS

1. **A pupil walks 24 blocks east. Which compass direction is the pupil moving?**
   A) south B) west C) east D) north
2. **A pupil walks 12 blocks north. Which compass direction is the pupil moving?**
   A) north B) east C) south D) west
3. **A pupil walks 4 blocks west. Which compass direction is the pupil moving?**
   A) west B) south C) east D) north
4. **A pupil walks 25 blocks south. Which compass direction is the pupil moving?**
   A) west B) east C) south D) north
5. **Ama faces east. Which direction is directly behind her?**
   A) east B) north C) south D) west

CHALLENGE · DIFFERENT EXAMPLES

1. **A learner walks toward north to school. To return on the same path, which direction must she walk?**
   A) west B) south C) east D) north
2. **Ama faces south. Which direction is directly behind her?**
   A) north B) east C) west D) south
3. **A scout faces east. After turning 90° clockwise, which direction does the scout face?**
   A) south B) east C) north D) west
4. **A pupil walks 14 blocks west. Which compass direction is the pupil moving?**
   A) north B) east C) south D) west
5. **A learner walks toward east to school. To return on the same path, which direction must she walk?**
   A) east B) south C) north D) west

Compass directions

CHALLENGE · APPLY WHAT YOU KNOW

6. **A pupil walks 9 blocks west. Which compass direction is the pupil moving?**
   A) east B) south C) north D) west

7. **A scout faces south. After turning 90° clockwise, which direction does the scout face?**
   A) south B) west C) north D) east

8. **Ama faces north. Which direction is directly behind her?**
   A) north B) east C) south D) west

9. **A pupil walks 6 blocks north. Which compass direction is the pupil moving?**
   A) east B) west C) south D) north

10. **A pupil walks 18 blocks west. Which compass direction is the pupil moving?**
    A) east B) south C) west D) north

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. east	2. north	3. west
4. south	5. west	6. south
7. north	8. south	9. west
10. west	11. west	12. west
13. south	14. north	15. west

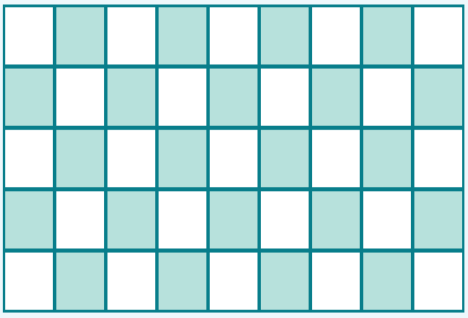
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Estimating perimeter & Measuring perimeter

LEARNING GOAL Estimating perimeter / Measuring perimeter

MATHS MODEL · GEOMETRY AND MEASUREMENT



9 columns × 5 rows = 45 square units

LEARN THE IDEA

Learn to estimate perimeter using referents measured in centimetres or metres and measure and record perimeter for regular and irregular shapes in cm and m.

WORKED EXAMPLE 1

Estimate the perimeter of a rectangle about 4 cm long and 4 cm wide using the side lengths.

Answer: 16 cm

Add all four sides once: 2 × (length + width).

WORKED EXAMPLE 2

Add the four outer sides of a 25 cm by 3 cm rectangular drawing. What is the perimeter?

Answer: 56 cm

25+3+25+3=56 cm.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Estimating perimeter & Measuring perimeter

PLAY · LEARN THE STEPS

1. **Estimate the perimeter of a rectangle about 7 cm long and 12 cm wide using the side lengths.**
   A) 84 cm B) 19 cm C) 38 cm D) 40 cm
2. **Add the four outer sides of a 11 cm by 8 cm rectangular drawing. What is the perimeter?**
   A) 19 cm B) 42 cm C) 38 cm D) 88 cm
3. **Estimate the perimeter of a rectangle about 11 cm long and 12 cm wide using the side lengths.**
   A) 46 cm B) 132 cm C) 48 cm D) 23 cm
4. **Add the four outer sides of a 3 cm by 13 cm rectangular drawing. What is the perimeter?**
   A) 32 cm B) 16 cm C) 39 cm D) 36 cm
5. **Estimate the perimeter of a rectangle about 8 cm long and 6 cm wide using the side lengths.**
   A) 28 cm B) 30 cm C) 48 cm D) 14 cm

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangular school garden measures 13 m by 3 m. How much fencing is needed for all four sides?**
   A) 39 m B) 34 m C) 32 m D) 16 m
2. **Estimate the perimeter of a rectangle about 9 cm long and 15 cm wide using the side lengths.**
   A) 50 cm B) 135 cm C) 48 cm D) 24 cm
3. **A rectangular school garden measures 15 m by 5 m. How much fencing is needed for all four sides?**
   A) 40 m B) 75 m C) 42 m D) 20 m
4. **Estimate the perimeter of a rectangle about 17 cm long and 10 cm wide using the side lengths.**
   A) 27 cm B) 54 cm C) 170 cm D) 56 cm
5. **A rectangular school garden measures 11 m by 9 m. How much fencing is needed for all four sides?**
   A) 42 m B) 99 m C) 40 m D) 20 m

Estimating perimeter & Measuring perimeter

CHALLENGE · APPLY WHAT YOU KNOW

6 **A rectangular school garden measures 16 m by 8 m. How much fencing is needed for all four sides?**

A) 128 m B) 24 m C) 50 m D) 48 m

7 **Add the four outer sides of a 11 cm by 24 cm rectangular drawing. What is the perimeter?**

A) 264 cm B) 35 cm C) 74 cm D) 70 cm

8 **A rectangular school garden measures 10 m by 2 m. How much fencing is needed for all four sides?**

A) 20 m B) 26 m C) 24 m D) 12 m

9 **Add the four outer sides of a 20 cm by 12 cm rectangular drawing. What is the perimeter?**

A) 32 cm B) 240 cm C) 68 cm D) 64 cm

10 **A rectangular school garden measures 9 m by 4 m. How much fencing is needed for all four sides?**

A) 26 m B) 28 m C) 13 m D) 36 m

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 38 cm	2. 38 cm	3. 46 cm
4. 32 cm	5. 28 cm	6. 32 m
7. 48 cm	8. 40 m	9. 54 cm
10. 40 m	11. 48 m	12. 70 cm
13. 24 m	14. 64 cm	15. 26 m

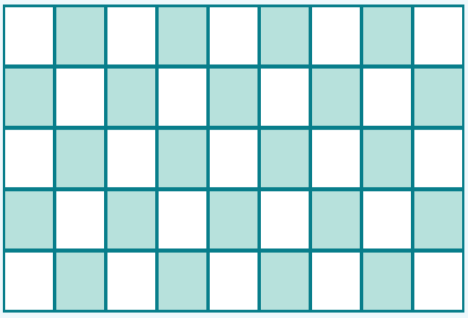
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Perimeter formulas & Same perimeter, different rectangles

LEARNING GOAL Perimeter formulas / Same perimeter, different rectangles

MATHS MODEL · GEOMETRY AND MEASUREMENT



9 columns × 5 rows = 45 square units

LEARN THE IDEA

Learn to develop and apply a formula for determining perimeter of square and rectangle and construct different rectangles for a given perimeter (cm, m) to demonstrate that many shapes are possible for a perimeter.

WORKED EXAMPLE 1

Calculate the perimeter of a 4 m long, 4 m wide school garden.

Answer: 16 m

P=2×(4+4)=16 m.

WORKED EXAMPLE 2

A rectangle has perimeter 64 cm and one side is 7 cm. How long is the other side?

Answer: 25 cm

Half the perimeter is 32; subtract 7.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Perimeter formulas & Same perimeter, different rectangles

PLAY · LEARN THE STEPS

1. **Calculate the perimeter of a 7 m long, 10 m wide school garden.**
   A) 70 m B) 17 m C) 34 m D) 32 m
2. **A rectangle has perimeter 66 cm and one side is 8 cm. How long is the other side?**
   A) 27 cm B) 33 cm C) 25 cm
3. **Calculate the perimeter of a 14 m long, 4 m wide school garden.**
   A) 36 m B) 34 m C) 56 m D) 18 m
4. **A rectangle has perimeter 66 cm and one side is 14 cm. How long is the other side?**
   A) 21 cm B) 19 cm C) 33 cm
5. **Calculate the perimeter of a 25 m long, 5 m wide school garden.**
   A) 30 m B) 58 m C) 60 m D) 125 m

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangular school garden measures 15 m by 4 m. How much fencing is needed for all four sides?**
   A) 60 m B) 38 m C) 40 m D) 19 m
2. **Calculate the perimeter of a 11 m long, 7 m wide school garden.**
   A) 18 m B) 77 m C) 34 m D) 36 m
3. **A rectangular school garden measures 8 m by 8 m. How much fencing is needed for all four sides?**
   A) 64 m B) 34 m C) 32 m D) 16 m
4. **A rectangular school garden measures 19 m by 2 m. How much fencing is needed for all four sides?**
   A) 44 m B) 42 m C) 21 m D) 38 m
5. **A rectangular school garden measures 9 m by 8 m. How much fencing is needed for all four sides?**
   A) 34 m B) 17 m C) 72 m D) 36 m

Perimeter formulas & Same perimeter, different rectangles

CHALLENGE · APPLY WHAT YOU KNOW

6. **A rectangular photo has perimeter 24 cm and width 4 cm. Find its length.**
A) 8 cm B) 12 cm C) 4 cm D) not 8 cm

7. **A rectangle has perimeter 46 cm and one side is 5 cm. How long is the other side?**
A) 18 cm B) 20 cm C) 23 cm

8. **A rectangular photo has perimeter 46 cm and width 9 cm. Find its length.**
A) 23 cm B) 14 cm C) 9 cm D) not 14 cm

9. **A rectangle has perimeter 56 cm and one side is 13 cm. How long is the other side?**
A) 15 cm B) 28 cm C) 17 cm

10. **A rectangular photo has perimeter 42 cm and width 8 cm. Find its length.**
A) 21 cm B) 8 cm C) not 13 cm D) 13 cm

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 34 m	2. 25 cm	3. 36 m
4. 19 cm	5. 60 m	6. 38 m
7. 36 m	8. 32 m	9. 42 m
10. 34 m	11. 8 cm	12. 18 cm
13. 14 cm	14. 15 cm	15. 13 cm

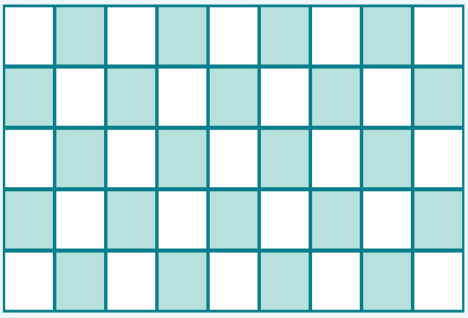
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Understanding area & Choosing area units

LEARNING GOAL Understanding area / Choosing area units

MATHS MODEL · GEOMETRY AND MEASUREMENT



9 columns × 5 rows = 45 square units

LEARN THE IDEA

Learn to recognise that area is measured in square units and select and justify referents for the units cm2 or m2.

WORKED EXAMPLE 1

A floor is covered by 3 rows of 2 equal square tiles. How many square units of area does it cover?

Answer: 6

Area means count all 6 unit squares.

WORKED EXAMPLE 2

An exercise book cover is 2 cm wide and 5 cm long. What is its area in square centimetres?

Answer: 10 cm²

A cover uses cm²; multiply its length and width.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Understanding area & Choosing area units

PLAY · LEARN THE STEPS

1. **A floor is covered by 5 rows of 12 equal square tiles. How many square units of area does it cover?**
   A) 110 B) 60 C) 80 D) 65
2. **An exercise book cover is 16 cm wide and 3 cm long. What is its area in square centimetres?**
   A) 48 cm² B) 48 m² C) 19 cm² D) 38 cm²
3. **A floor is covered by 7 rows of 5 equal square tiles. How many square units of area does it cover?**
   A) 235 B) 35 C) 31 D) 33
4. **An exercise book cover is 11 cm wide and 12 cm long. What is its area in square centimetres?**
   A) 23 cm² B) 132 cm² C) 46 cm² D) 132 m²
5. **A floor is covered by 13 rows of 6 equal square tiles. How many square units of area does it cover?**
   A) 68 B) 74 C) 76 D) 78

CHALLENGE · DIFFERENT EXAMPLES

1. **A display mat has 14 rows with 3 square tiles each. How many square units are covered?**
   A) 17 B) 42 C) 45 D) 34
2. **A display mat has 16 rows with 6 square tiles each. How many square units are covered?**
   A) 22 B) 102 C) 96 D) 44
3. **A floor is covered by 9 rows of 14 equal square tiles. How many square units of area does it cover?**
   A) 226 B) 176 C) 126 D) 128
4. **A display mat has 4 rows with 2 square tiles each. How many square units are covered?**
   A) 10 B) 8 C) 6 D) 12
5. **A display mat has 8 rows with 6 square tiles each. How many square units are covered?**
   A) 48 B) 54 C) 28 D) 14

Understanding area & Choosing area units

CHALLENGE · APPLY WHAT YOU KNOW

6 **A display mat has 4 rows with 4 square tiles each. How many square units are covered?**

A) 20 B) 16 C) 8 D) 18

7 **An exercise book cover is 7 cm wide and 6 cm long. What is its area in square centimetres?**

A) 26 cm² B) 42 cm² C) 42 m² D) 13 cm²

8 **A display mat has 14 rows with 7 square tiles each. How many square units are covered?**

A) 98 B) 105 C) 21 D) 42

9 **An exercise book cover is 2 cm wide and 10 cm long. What is its area in square centimetres?**

A) 20 m² B) 24 cm² C) 20 cm² D) 12 cm²

10 **A display mat has 11 rows with 7 square tiles each. How many square units are covered?**

A) 36 B) 84 C) 18 D) 77

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 60	2. 48 cm²	3. 35
4. 132 cm²	5. 78	6. 42
7. 96	8. 126	9. 8
10. 48	11. 16	12. 42 cm²
13. 98	14. 20 cm²	15. 77

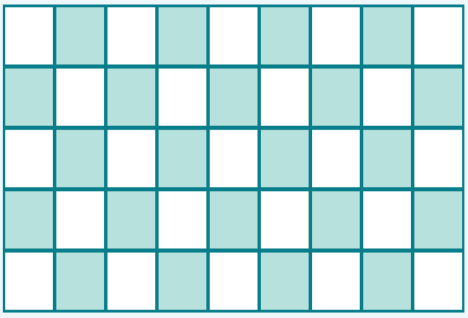
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Estimating area & Area formulas

LEARNING GOAL Estimating area / Area formulas

MATHS MODEL · GEOMETRY AND MEASUREMENT



9 columns × 5 rows = 45 square units

LEARN THE IDEA

Learn to estimate area by using referents for cm2 or m2 and develop and apply a formula for determining area of a rectangle and square.

WORKED EXAMPLE 1

Estimate the number of 1 cm² tiles covering a 3 cm by 2 cm rectangle.

Answer: 6

Each unit square contributes 1 cm².

WORKED EXAMPLE 2

What is the area of a rectangle 2 cm by 5 cm?

Answer: 10 cm²

Area = length × width = 10 cm².

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Estimating area & Area formulas

PLAY · LEARN THE STEPS

1. **Estimate the number of 1 cm² tiles covering a 10 cm by 7 cm rectangle.**
   A) 1070 B) 70 C) 74 D) 69
2. **What is the area of a rectangle 2 cm by 10 cm?**
   A) 22 cm² B) 12 cm² C) 24 cm² D) 20 cm²
3. **Estimate the number of 1 cm² tiles covering a 15 cm by 14 cm rectangle.**
   A) 208 B) 214 C) 260 D) 210
4. **What is the area of a rectangle 5 cm by 7 cm?**
   A) 37 cm² B) 12 cm² C) 24 cm² D) 35 cm²
5. **Estimate the number of 1 cm² tiles covering a 12 cm by 11 cm rectangle.**
   A) 132 B) 332 C) 182 D) 129

CHALLENGE · DIFFERENT EXAMPLES

1. **A display mat has 9 rows with 2 square tiles each. How many square units are covered?**
   A) 20 B) 18 C) 22 D) 11
2. **Estimate the number of 1 cm² tiles covering a 16 cm by 14 cm rectangle.**
   A) 244 B) 234 C) 174 D) 224
3. **A display mat has 16 rows with 9 square tiles each. How many square units are covered?**
   A) 153 B) 25 C) 50 D) 144
4. **Estimate the number of 1 cm² tiles covering a 4 cm by 10 cm rectangle.**
   A) 140 B) 40 C) 44 D) 43
5. **A display mat has 3 rows with 5 square tiles each. How many square units are covered?**
   A) 20 B) 16 C) 15 D) 8

Estimating area & Area formulas

CHALLENGE · APPLY WHAT YOU KNOW

6 **A display mat has 7 rows with 4 square tiles each. How many square units are covered?**

A) 22 B) 32 C) 11 D) 28

7 **What is the area of a rectangle 4 cm by 13 cm?**

A) 52 cm² B) 54 cm² C) 34 cm² D) 17 cm²

8 **A display mat has 5 rows with 3 square tiles each. How many square units are covered?**

A) 18 B) 16 C) 8 D) 15

9 **What is the area of a rectangle 14 cm by 3 cm?**

A) 42 cm² B) 17 cm² C) 34 cm² D) 44 cm²

10 **A display mat has 12 rows with 5 square tiles each. How many square units are covered?**

A) 34 B) 65 C) 60 D) 17

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 70	2. 20 cm²	3. 210
4. 35 cm²	5. 132	6. 18
7. 224	8. 144	9. 40
10. 15	11. 28	12. 52 cm²
13. 15	14. 42 cm²	15. 60

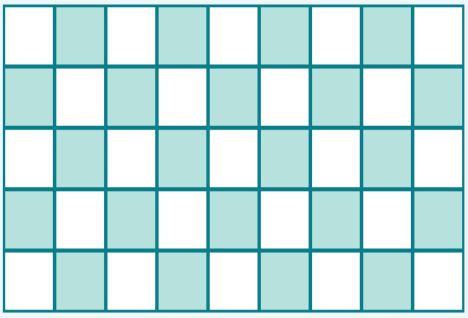
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Same area, different rectangles

LEARNING GOAL Same area, different rectangles

MATHS MODEL · GEOMETRY AND MEASUREMENT



9 columns × 5 rows = 45 square units

LEARN THE IDEA

Learn to construct different rectangles for a given area (cm2 or m2) in order to demonstrate that many different rectangles may have the same area.

WORKED EXAMPLE 1

A rectangle has area 21 cm² and length 3 cm. What is the width?

Answer: 7 cm

21 ÷ 3 = 7.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Same area, different rectangles

PLAY · LEARN THE STEPS

1. **A rectangle has area 10 cm² and length 2 cm. What is the width?**
   A) 5 cm B) 12 cm C) 4 cm D) 7 cm
2. **A rectangle has area 12 cm² and length 3 cm. What is the width?**
   A) 3 cm B) 15 cm C) 4 cm D) 7 cm
3. **A rectangle has area 168 cm² and length 14 cm. What is the width?**
   A) 12 cm B) 11 cm C) 26 cm D) 182 cm
4. **A rectangle has area 154 cm² and length 14 cm. What is the width?**
   A) 25 cm B) 168 cm C) 11 cm D) 10 cm
5. **A rectangle has area 162 cm² and length 18 cm. What is the width?**
   A) 8 cm B) 180 cm C) 9 cm D) 27 cm

CHALLENGE · DIFFERENT EXAMPLES

1. **A rectangular plot is 3 m by 3 m. What total area does the plot cover?**
   A) 9 m² total B) 10 m² total C) 12 m² total D) 6 m² total
2. **A rectangle has area 105 cm² and length 15 cm. What is the width?**
   A) 120 cm B) 22 cm C) 6 cm D) 7 cm
3. **A play area covers 90 square metres and is 15 m long. How wide is it?**
   A) 6 m B) 105 m C) 84 m D) 15 m
4. **A rectangle has area 240 cm² and length 16 cm. What is the width?**
   A) 15 cm B) 14 cm C) 31 cm D) 256 cm
5. **A rectangular plot is 8 m by 4 m. What total area does the plot cover?**
   A) 32 m² total B) 24 m² total C) 36 m² total D) 33 m² total

Same area, different rectangles

CHALLENGE · APPLY WHAT YOU KNOW

6. **A rectangle has area 32 cm² and length 16 cm. What is the width?**
   A) 1 cm B) 2 cm C) 18 cm D) 48 cm

7. **A play area covers 32 square metres and is 8 m long. How wide is it?**
   A) 28 m B) 8 m C) 4 m D) 40 m

8. **A rectangle has area 25 cm² and length 5 cm. What is the width?**
   A) 30 cm B) 4 cm C) 5 cm D) 10 cm

9. **A rectangular plot is 5 m by 9 m. What total area does the plot cover?**
   A) 45 m² total B) 46 m² total C) 40 m² total D) 54 m² total

10. **A rectangle has area 33 cm² and length 3 cm. What is the width?**
    A) 14 cm B) 11 cm C) 10 cm D) 36 cm

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 5 cm	2. 4 cm	3. 12 cm
4. 11 cm	5. 9 cm	6. 9 m² total
7. 7 cm	8. 6 m	9. 15 cm
10. 32 m² total	11. 2 cm	12. 4 m
13. 5 cm	14. 45 m² total	15. 11 cm

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Clock times & Timing events

LEARNING GOAL Clock times / Timing events

MATHS MODEL · GEOMETRY AND MEASUREMENT



LEARN THE IDEA

Learn to tell the time in hours and minutes in analogue and digital watches including 24-hour clocks and use clock to measure time to complete simple events in minutes and seconds.

WORKED EXAMPLE 1

Write 01:00 in 24-hour time if it is in the afternoon.

Answer: 13:00

For PM hours 1–11, add 12 to the hour.

WORKED EXAMPLE 2

A timer reads 24 minutes 5 seconds. How many seconds is that?

Answer: 1445

24×60+5=1445 seconds.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Clock times & Timing events

PLAY · LEARN THE STEPS

1. **Write 12:00 in 24-hour time if it is in the afternoon.**
   A) 12:00 B) 11:00 C) 12:10
2. **A timer reads 22 minutes 43 seconds. How many seconds is that?**
   A) 1343 B) 1362 C) 1353 D) 1363
3. **Write 11:50 in 24-hour time if it is in the afternoon.**
   A) 23:00 B) 22:50 C) 11:50 D) 23:50
4. **A timer reads 11 minutes 10 seconds. How many seconds is that?**
   A) 674 B) 670 C) 680 D) 770
5. **Write 07:10 in 24-hour time if it is in the afternoon.**
   A) 18:10 B) 19:20 C) 07:10 D) 19:10

CHALLENGE · DIFFERENT EXAMPLES

1. **A lesson begins at 2:15 PM. What time is that on a 24-hour clock?**
   A) 14:30 B) 02:15 C) 14:15 D) 13:15
2. **Write 03:50 in 24-hour time if it is in the afternoon.**
   A) 15:50 B) 03:50 C) 15:00 D) 14:50
3. **A lesson begins at 8:15 PM. What time is that on a 24-hour clock?**
   A) 20:30 B) 19:15 C) 20:15 D) 08:15
4. **Write 04:20 in 24-hour time if it is in the afternoon.**
   A) 16:20 B) 15:20 C) 16:30 D) 04:20
5. **A lesson begins at 9:15 PM. What time is that on a 24-hour clock?**
   A) 21:15 B) 20:15 C) 21:30 D) 09:15

Clock times & Timing events

CHALLENGE · APPLY WHAT YOU KNOW

6. **A race takes 23 minutes and 4 seconds. How many seconds is that?**
   A) 1444 B) 1380 C) 27 D) 1384

7. **A timer reads 24 minutes 19 seconds. How many seconds is that?**
   A) 1459 B) 1463 C) 1469 D) 1559

8. **A race takes 17 minutes and 25 seconds. How many seconds is that?**
   A) 1020 B) 42 C) 1105 D) 1045

9. **A timer reads 2 minutes 29 seconds. How many seconds is that?**
   A) 129 B) 149 C) 153 D) 169

10. **A race takes 19 minutes and 48 seconds. How many seconds is that?**
    A) 1140 B) 67 C) 1248 D) 1188

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 12:00	2. 1363	3. 23:50
4. 670	5. 19:10	6. 14:15
7. 15:50	8. 20:15	9. 16:20
10. 21:15	11. 1384	12. 1459
13. 1045	14. 149	15. 1188

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Writing calendar dates

LEARNING GOAL Writing calendar dates

MATHS MODEL · GEOMETRY AND MEASUREMENT



LEARN THE IDEA

Learn to state dates of events and record calendar dates in a variety of formats.

WORKED EXAMPLE 1

Write 2 January 2026 as DD/MM/YYYY.

Answer: 02/01/2026

Day first, then month, then four-digit year.

TRY THIS IN REAL LIFE

Use paper, a ruler and a pencil to draw the shape or measure a real object. Compare an estimate with a measurement.

Writing calendar dates

PLAY · LEARN THE STEPS

1. **Write 24 December 2026 as DD/MM/YYYY.**
   A) 24/12/2027 B) 12/24/2026 C) 24/12/2025 D) 24/12/2026
2. **Write 2 July 2028 as DD/MM/YYYY.**
   A) 02/07/2028 B) 02/07/2029 C) 07/02/2028 D) 02/07/2027
3. **Write 7 February 2025 as DD/MM/YYYY.**
   A) 02/07/2025 B) 07/02/2025 C) 07/02/2026 D) 07/02/2024
4. **Write 2 April 2028 as DD/MM/YYYY.**
   A) 04/02/2028 B) 02/04/2029 C) 02/04/2028 D) 02/04/2027
5. **Write 25 August 2024 as DD/MM/YYYY.**
   A) 25/08/2024 B) 08/25/2024 C) 25/08/2023 D) 25/08/2025

CHALLENGE · DIFFERENT EXAMPLES

1. **A school fair is on 13/03/2028. What is the date 1 day later?**
   A) 14/03/2028 B) 12/03/2028 C) 15/03/2028 D) 17/03/2028
2. **Write 18 January 2028 as DD/MM/YYYY.**
   A) 18/01/2028 B) 18/01/2029 C) 18/01/2027 D) 01/18/2028
3. **A notice says September 14, 2028. Write the date as DD/MM/YYYY.**
   A) 14/09/2028 B) 14/09/2027 C) 09/14/2028 D) 14/09/2029
4. **A school fair is on 22/04/2026. What is the date 1 day later?**
   A) 21/04/2026 B) 24/04/2026 C) 23/04/2026 D) not 23/04/2026
5. **Write 23 August 2027 as DD/MM/YYYY.**
   A) 23/08/2027 B) 23/08/2028 C) 08/23/2027 D) 23/08/2026

Writing calendar dates

CHALLENGE · APPLY WHAT YOU KNOW

6. **A notice says November 14, 2028. Write the date as DD/MM/YYYY.**
A) 14/11/2028 B) 11/14/2028 C) 14/11/2029 D) 14/11/2027

7. **Write 7 June 2027 as DD/MM/YYYY.**
A) 06/07/2027 B) 07/06/2028 C) 07/06/2027 D) 07/06/2026

8. **A school fair is on 03/01/2027. What is the date 1 day later?**
A) 06/01/2027 B) 04/01/2027 C) 05/01/2027 D) 02/01/2027

9. **Write 6 June 2027 as DD/MM/YYYY.**
A) 06/06/2027 B) 06/06/2028 C) 06/06/2026

10. **A notice says October 2, 2029. Write the date as DD/MM/YYYY.**
A) 10/02/2029 B) 02/10/2029 C) 02/10/2028 D) 02/10/2030

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 24/12/2026	2. 02/07/2028	3. 07/02/2025
4. 02/04/2028	5. 25/08/2024	6. 14/03/2028
7. 18/01/2028	8. 14/09/2028	9. 23/04/2026
10. 23/08/2027	11. 14/11/2028	12. 07/06/2027
13. 04/01/2027	14. 06/06/2027	15. 02/10/2029

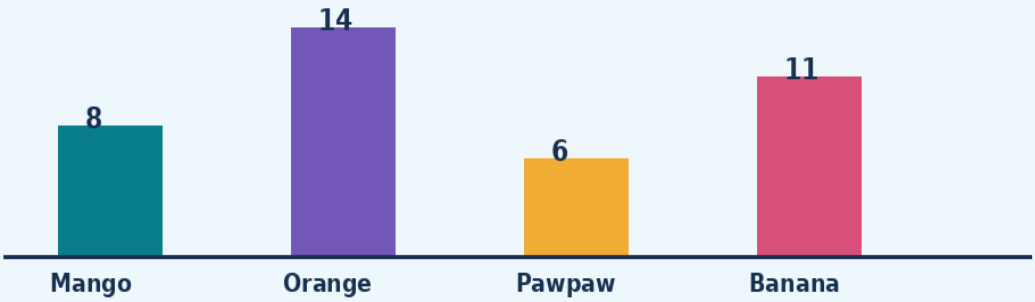
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

One symbol, one item & Pictograph keys

LEARNING GOAL One symbol, one item / Pictograph keys

MATHS MODEL · DATA AND GRAPHS



LEARN THE IDEA

Learn to use an understanding of one-to-one correspondence to read and interpret graphs and use an understanding of many-to-one correspondence to display or construct graphs.

WORKED EXAMPLE 1

In a chart, one picture means one pupil. There are 13 pawpaw pictures. How many pupils chose pawpaw?

Answer: 13

One symbol represents one pupil.

WORKED EXAMPLE 2

Each pictograph symbol represents 2 fruits. How many fruits do 9 symbols represent?

Answer: 18

9×2=18.

TRY THIS IN REAL LIFE

Ask four classmates about their favourite local fruits. Make a tally, then draw a labelled bar graph with an accurate scale.

One symbol, one item & Pictograph keys

PLAY · LEARN THE STEPS

1. **In a chart, one picture means one pupil. There are 7 orange pictures. How many pupils chose orange?**
   A) 6 B) 10 C) 7 D) 8
2. **Each pictograph symbol represents 1 fruit. How many fruits do 16 symbols represent?**
   A) 16 B) 36 C) 12 D) 20
3. **In a chart, one picture means one pupil. There are 9 banana pictures. How many pupils chose banana?**
   A) 11 B) 9 C) 29 D) 8
4. **Each pictograph symbol represents 5 fruits. How many fruits do 10 symbols represent?**
   A) 50 B) 60 C) 47 D) 46
5. **In a chart, one picture means one pupil. There are 2 mango pictures. How many pupils chose mango?**
   A) 4 B) 6 C) 2 D) 1

CHALLENGE · DIFFERENT EXAMPLES

1. **A pictograph uses one picture for one pupil. There are 17 pictures of mangoes. How many pupils chose them?**
   A) 18 B) 34 C) 17 D) 16
2. **A pictograph uses one picture for one pupil. There are 14 pictures of oranges. How many pupils chose them?**
   A) 15 B) 14 C) 28 D) 13
3. **A pictograph uses one picture for one pupil. There are 7 pictures of pawpaws. How many pupils chose them?**
   A) 7 B) 8 C) 14 D) 6
4. **In a chart, one picture means one pupil. There are 17 mango pictures. How many pupils chose mango?**
   A) 27 B) 7 C) 17 D) 37
5. **A pictograph uses one picture for one pupil. There are 4 pictures of bananas. How many pupils chose them?**
   A) 5 B) 3 C) 4 D) 8

One symbol, one item & Pictograph keys

CHALLENGE · APPLY WHAT YOU KNOW

6 **Each fruit icon stands for 2 mangoes. What do 6 icons represent?**

A) 14 B) 12 C) 10 D) 6

7 **Each fruit icon stands for 1 oranges. What do 11 icons represent?**

A) 13 B) 11 C) 10 D) 12

8 **Each pictograph symbol represents 1 fruit. How many fruits do 4 symbols represent?**

A) 2 B) 0 C) 1 D) 4

9 **Each fruit icon stands for 1 pawpaws. What do 9 icons represent?**

A) 11 B) 8 C) 9 D) 10

10 **Each pictograph symbol represents 2 fruits. How many fruits do 5 symbols represent?**

A) 7 B) 1010 C) 10 D) 11

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 7	2. 16	3. 9
4. 50	5. 2	6. 17
7. 14	8. 7	9. 17
10. 4	11. 12	12. 11
13. 4	14. 9	15. 10

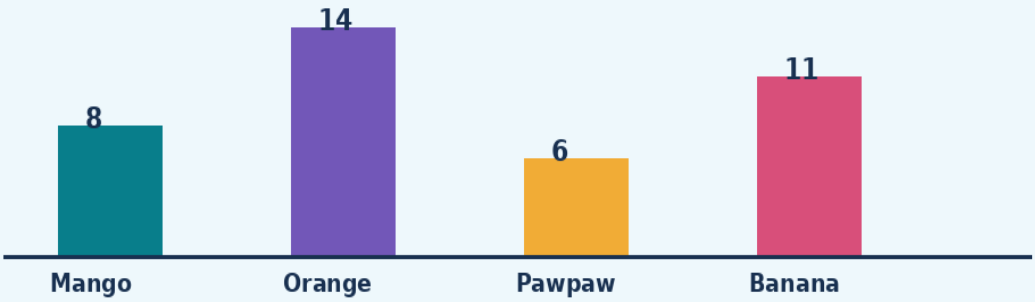
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Comparing charts & Graphs in everyday life

LEARNING GOAL Comparing charts / Graphs in everyday life

MATHS MODEL · DATA AND GRAPHS



LEARN THE IDEA

Learn to compare graphs in which the same data has been displayed and explain how they are the same and different and find examples of graphs in which many-to-one correspondence is used in print and electronic media, such as newspapers, magazines and the internet, and describe the correspondence used.

WORKED EXAMPLE 1

A table lists pawpaw: 13 and mango: 3. How many more are in the larger group?

Answer: 10

Subtract the smaller bar height from the larger.

WORKED EXAMPLE 2

A pictograph key says 1 symbol = 2 items. How many symbols show 18 items?

Answer: 9

Divide the item total by the key value.

TRY THIS IN REAL LIFE

Ask four classmates about their favourite local fruits. Make a tally, then draw a labelled bar graph with an accurate scale.

Comparing charts & Graphs in everyday life

PLAY · LEARN THE STEPS

1. **A table lists orange: 26 and banana: 8. How many more are in the larger group?**
   A) 21 B) 28 C) 16 D) 18
2. **A pictograph key says 1 symbol = 1 item. How many symbols show 3 items?**
   A) 3 B) 8 C) 13 D) 5
3. **A table lists mango: 9 and pawpaw: 10. How many more are in the larger group?**
   A) 1 B) 1001 C) 51 D) 5
4. **A pictograph key says 1 symbol = 5 items. How many symbols show 10 items?**
   A) 5 B) 52 C) 6 D) 2
5. **A table lists banana: 13 and orange: 13. How many more are in the larger group?**
   A) 10 B) 5 C) 50 D) 0

CHALLENGE · DIFFERENT EXAMPLES

1. **A bar chart shows mangoes: 24 and bananas: 11. What is the difference between their bars?**
   A) 15 B) 24 C) 13 D) 35
2. **A table lists mango: 12 and pawpaw: 14. How many more are in the larger group?**
   A) 2 B) 1002 C) 102 D) 1
3. **A bar chart shows oranges: 12 and bananas: 16. What is the difference between their bars?**
   A) 6 B) 4 C) 12 D) 28
4. **A table lists orange: 32 and banana: 4. How many more are in the larger group?**
   A) 78 B) 28 C) 25 D) 1028
5. **A bar chart shows pawpaws: 45 and bananas: 12. What is the difference between their bars?**
   A) 57 B) 35 C) 45 D) 33

Comparing charts & Graphs in everyday life

CHALLENGE · APPLY WHAT YOU KNOW

6 **Each graph icon stands for 1 votes. How many icons are needed to show 17 votes?**

A) 16 B) 17 C) 19 D) 18

7 **A pictograph key says 1 symbol = 5 items. How many symbols show 25 items?**

A) 205 B) 9 C) 5 D) 10

8 **Each graph icon stands for 2 votes. How many icons are needed to show 16 votes?**

A) 16 B) 10 C) 8 D) 6

9 **Each graph icon stands for 1 votes. How many icons are needed to show 11 votes?**

A) 11 B) 13 C) 12 D) 10

10 **A pictograph key says 1 symbol = 1 item. How many symbols show 12 items?**

A) 12 B) 112 C) 1012 D) 14

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 18	2. 3	3. 1
4. 2	5. 0	6. 13
7. 2	8. 4	9. 28
10. 33	11. 17	12. 5
13. 8	14. 11	15. 12

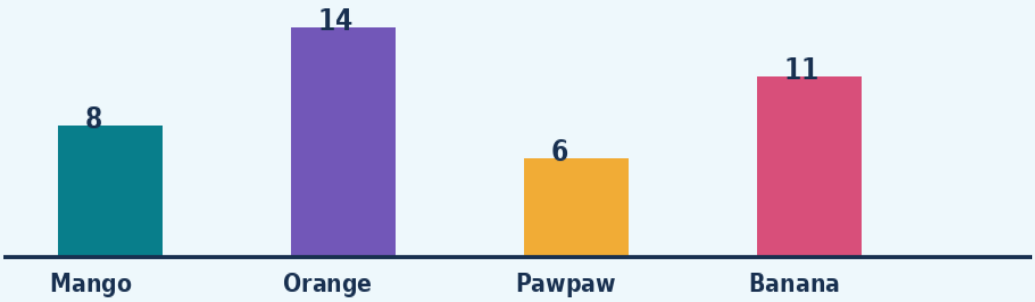
REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.

Create a graph & Read graphs to decide

LEARNING GOAL Create a graph / Read graphs to decide

MATHS MODEL · DATA AND GRAPHS



LEARN THE IDEA

Learn to identify common features of graphs that use many-to-one correspondence and use that understanding to draw bar graphs or pictographs, complete with title, labelled axes, key or legend, to represent data collected (up to 4 categories of data) and use understanding of many-to-one …

WORKED EXAMPLE 1

A bar chart displays pawpaw=13. Which bar height should be used?

Answer: 13

The labelled vertical axis must reach the recorded value.

WORKED EXAMPLE 2

A fruit chart shows banana=18 and orange=18. How many altogether?

Answer: 36

Add the two category totals: 18+18.

TRY THIS IN REAL LIFE

Ask four classmates about their favourite local fruits. Make a tally, then draw a labelled bar graph with an accurate scale.

Create a graph & Read graphs to decide

PLAY · LEARN THE STEPS

1. **A bar chart displays banana=36. Which bar height should be used?**
   A) 36 B) 41 C) 26 D) 32
2. **A fruit chart shows pawpaw=11 and mango=9. How many altogether?**
   A) 1020 B) 20 C) 19 D) 40
3. **A bar chart displays mango=10. Which bar height should be used?**
   A) 6 B) 8 C) 13 D) 10
4. **A fruit chart shows orange=10 and banana=50. How many altogether?**
   A) 70 B) 59 C) 60 D) 64
5. **A bar chart displays pawpaw=25. Which bar height should be used?**
   A) 35 B) 25 C) 225 D) 5

CHALLENGE · DIFFERENT EXAMPLES

1. **A class recorded 4 votes for mangoes. How high should its bar be on an axis marked in ones?**
   A) 4 B) 9 C) 3 D) 6
2. **A bar chart displays banana=22. Which bar height should be used?**
   A) 222 B) 26 C) 2 D) 22
3. **A class recorded 4 votes for oranges. How high should its bar be on an axis marked in ones?**
   A) 9 B) 4 C) 6 D) 3
4. **A bar chart displays pawpaw=30. Which bar height should be used?**
   A) 28 B) 80 C) 30 D) 25
5. **A class recorded 12 votes for pawpaws. How high should its bar be on an axis marked in ones?**
   A) 17 B) 11 C) 14 D) 12

Create a graph & Read graphs to decide

CHALLENGE · APPLY WHAT YOU KNOW

6 **The mangoes bar is 35 and the orange bar is 16. How many votes were recorded for both?**

A) 53 B) 35 C) 16 D) 51

7 **A fruit chart shows pawpaw=14 and mango=18. How many altogether?**

A) 22 B) 33 C) 28 D) 32

8 **The oranges bar is 55 and the orange bar is 6. How many votes were recorded for both?**

A) 61 B) 55 C) 6 D) 63

9 **A fruit chart shows mango=10 and pawpaw=2. How many altogether?**

A) 112 B) 212 C) 12 D) 1012

10 **The pawpaws bar is 70 and the orange bar is 12. How many votes were recorded for both?**

A) 82 B) 12 C) 70 D) 84

ANSWER CHECK · LOOK ONLY AFTER TRYING

1. 36	2. 20	3. 10
4. 60	5. 25	6. 4
7. 22	8. 4	9. 30
10. 12	11. 51	12. 32
13. 61	14. 12	15. 82

REFLECT

Which Challenge used a different method from Play? Explain your reasoning to a partner.